

```
[terraform-user@ip-172-31-5-142 terraform-project]$ vi provider.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ vi vpc.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ vi vpc.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ vi vpc.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform init
Initializing provider plugins found in the configuration...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.51.0
```

Initializing the backend...

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform validate
Success! The configuration is valid.
```

```
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform plan
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```
# aws_internet_gateway.igw will be created
+ resource "aws_internet_gateway" "igw" {
  + arn      = (known after apply)
  + id       = (known after apply)
  + owner_id = (known after apply)
  + region   = "ap-south-2"
  + tags     = {
    + "Name" = "terraform-igw"
  }
  + tags_all = {
    + "Name" = "terraform-igw"
  }
  + vpc_id   = (known after apply)
}

# aws_route_table.public will be created
+ resource "aws_route_table" "public" {
  + arn              = (known after apply)
  + id               = (known after apply)
  + owner_id         = (known after apply)
  + propagating_vgws = (known after apply)
  + region           = "ap-south-2"
```

```

+ route          = [
+   {
+     + cidr_block          = "0.0.0.0/0"
+     + gateway_id          = (known after apply)
+     # (12 unchanged attributes hidden)
+   },
+ ]
+ tags            = {
+   + "Name" = "public-rt"
+ }
+ tags_all        = {
+   + "Name" = "public-rt"
+ }
+ vpc_id          = (known after apply)
}

# aws_route_table_association.public will be created
+ resource "aws_route_table_association" "public" {
+   id            = (known after apply)
+   region        = "ap-south-2"
+   route_table_id = (known after apply)
+   subnet_id     = (known after apply)
+ }

# aws_subnet.public will be created
+ resource "aws_subnet" "public" {
+   arn                                = (known after apply)
+   assign_ipv6_address_on_creation    = false
+   availability_zone                 = "ap-south-2a"
+   availability_zone_id              = (known after apply)
+   cidr_block                        = "10.0.1.0/24"
+   enable_dns64                      = false
+   enable_resource_name_dns_a_record_on_launch = false
+   enable_resource_name_dns_aaaa_record_on_launch = false
+   id                                = (known after apply)
+   ipv6_cidr_block                   = (known after apply)
+   ipv6_cidr_block_association_id    = (known after apply)
+   ipv6_native                       = false
+   map_public_ip_on_launch           = true
+   owner_id                          = (known after apply)
+   private_dns_hostname_type_on_launch = (known after apply)
+   region                            = "ap-south-2"
+   tags                              = {
+     + "Name" = "public-subnet"
+   }
+   tags_all                          = {
+     + "Name" = "public-subnet"
+   }
+   vpc_id                            = (known after apply)
+ }

# aws_vpc.main will be created
+ resource "aws_vpc" "main" {
+   arn          = (known after apply)
+   cidr_block    = "10.0.0.0/16"

```

```

+ default_network_acl_id           = (known after apply)
+ default_route_table_id           = (known after apply)
+ default_security_group_id        = (known after apply)
+ dhcp_options_id                  = (known after apply)
+ enable_dns_hostnames              = (known after apply)
+ enable_dns_support                = true
+ enable_network_address_usage_metrics = (known after apply)
+ id                               = (known after apply)
+ instance_tenancy                  = "default"
+ ipv6_association_id               = (known after apply)
+ ipv6_cidr_block                   = (known after apply)
+ ipv6_cidr_block_network_border_group = (known after apply)
+ main_route_table_id               = (known after apply)
+ owner_id                          = (known after apply)
+ region                            = "ap-south-2"
+ tags                              = {
    + "Name" = "terraform-vpc"
  }
+ tags_all                          = {
    + "Name" = "terraform-vpc"
  }
}

```

Plan: 5 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

[terraform-user@ip-172-31-5-142 terraform-project]\$ terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

aws_internet_gateway.igw will be created

```

+ resource "aws_internet_gateway" "igw" {
  + arn      = (known after apply)
  + id       = (known after apply)
  + owner_id = (known after apply)
  + region   = "ap-south-2"
  + tags     = {
    + "Name" = "terraform-igw"
  }
  + tags_all = {
    + "Name" = "terraform-igw"
  }
  + vpc_id   = (known after apply)
}

```

aws_route_table.public will be created

```

+ resource "aws_route_table" "public" {

```

```

+ arn                = (known after apply)
+ id                 = (known after apply)
+ owner_id           = (known after apply)
+ propagating_vgws   = (known after apply)
+ region             = "ap-south-2"
+ route              = [
  + {
    + cidr_block      = "0.0.0.0/0"
    + gateway_id      = (known after apply)
    # (12 unchanged attributes hidden)
  },
]
+ tags               = {
  + "Name" = "public-rt"
}
+ tags_all           = {
  + "Name" = "public-rt"
}
+ vpc_id             = (known after apply)
}

# aws_route_table_association.public will be created
+ resource "aws_route_table_association" "public" {
  + id                = (known after apply)
  + region            = "ap-south-2"
  + route_table_id    = (known after apply)
  + subnet_id         = (known after apply)
}

# aws_subnet.public will be created
+ resource "aws_subnet" "public" {
  + arn                = (known after apply)
  + assign_ipv6_address_on_creation = false
  + availability_zone   = "ap-south-2a"
  + availability_zone_id = (known after apply)
  + cidr_block          = "10.0.1.0/24"
  + enable_dns64        = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                 = (known after apply)
  + ipv6_cidr_block     = (known after apply)
  + ipv6_cidr_block_association_id = (known after apply)
  + ipv6_native         = false
  + map_public_ip_on_launch = true
  + owner_id           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region             = "ap-south-2"
  + tags               = {
    + "Name" = "public-subnet"
  }
  + tags_all           = {
    + "Name" = "public-subnet"
  }
  + vpc_id             = (known after apply)
}

```

```
# aws_vpc.main will be created
+ resource "aws_vpc" "main" {
  + arn                                = (known after apply)
  + cidr_block                        = "10.0.0.0/16"
  + default_network_acl_id           = (known after apply)
  + default_route_table_id           = (known after apply)
  + default_security_group_id        = (known after apply)
  + dhcp_options_id                  = (known after apply)
  + enable_dns_hostnames              = (known after apply)
  + enable_dns_support                = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                                = (known after apply)
  + instance_tenancy                  = "default"
  + ipv6_association_id               = (known after apply)
  + ipv6_cidr_block                   = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id               = (known after apply)
  + owner_id                          = (known after apply)
  + region                            = "ap-south-2"
  + tags                              = {
    + "Name" = "terraform-vpc"
  }
  + tags_all                          = {
    + "Name" = "terraform-vpc"
  }
}
```

Plan: 5 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

aws_vpc.main: Creating...

aws_vpc.main: Creation complete after 1s [id=vpc-0ed3dccc5d799fbb9]

aws_internet_gateway.igw: Creating...

aws_subnet.public: Creating...

aws_internet_gateway.igw: Creation complete after 0s [id=igw-0ce5164dc047e0a65]

aws_route_table.public: Creating...

aws_route_table.public: Creation complete after 1s [id=rtb-0e058852b8dad8c2d]

aws_subnet.public: Still creating... [00m10s elapsed]

aws_subnet.public: Creation complete after 11s [id=subnet-0caf4afde5ceb4dce]

aws_route_table_association.public: Creating...

aws_route_table_association.public: Creation complete after 0s

[id=rtbassoc-04a813e5b7f3a67bf]

Apply complete! Resources: 5 added, 0 changed, 0 destroyed.

[terraform-user@ip-172-31-5-142 terraform-project]\$

[terraform-user@ip-172-31-5-142 terraform-project]\$ vi security.tf

[terraform-user@ip-172-31-5-142 terraform-project]\$ vi ec2.tf

[terraform-user@ip-172-31-5-142 terraform-project]\$ terraform fmt

[terraform-user@ip-172-31-5-142 terraform-project]\$ terraform validate

Success! The configuration is valid.

```
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform plan
aws_vpc.main: Refreshing state... [id=vpc-0ed3dccd5d799fbb9]
aws_subnet.public: Refreshing state... [id=subnet-0caf4afde5ceb4dce]
aws_internet_gateway.igw: Refreshing state... [id=igw-0ce5164dc047e0a65]
aws_route_table.public: Refreshing state... [id=rtb-0e058852b8dad8c2d]
aws_route_table_association.public: Refreshing state...
[id=rtbassoc-04a813e5b7f3a67bf]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```
# aws_instance.demo will be created
+ resource "aws_instance" "demo" {
  + ami                        = "ami-04a64102b8022e4f3"
  + arn                       = (known after apply)
  + associate_public_ip_address = true
  + availability_zone          = (known after apply)
  + disable_api_stop           = (known after apply)
  + disable_api_termination    = (known after apply)
  + ebs_optimized              = (known after apply)
  + enable_primary_ipv6        = (known after apply)
  + force_destroy              = false
  + get_password_data          = false
  + host_id                    = (known after apply)
  + host_resource_group_arn     = (known after apply)
  + iam_instance_profile        = (known after apply)
  + id                         = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle          = (known after apply)
  + instance_state              = (known after apply)
  + instance_type               = "t3.micro"
  + ipv6_address_count          = (known after apply)
  + ipv6_addresses              = (known after apply)
  + key_name                    = (known after apply)
  + monitoring                  = (known after apply)
  + outpost_arn                 = (known after apply)
  + password_data               = (known after apply)
  + placement_group             = (known after apply)
  + placement_group_id          = (known after apply)
  + placement_partition_number  = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns                 = (known after apply)
  + private_ip                  = (known after apply)
  + public_dns                  = (known after apply)
  + public_ip                   = (known after apply)
  + region                      = "ap-south-2"
  + secondary_private_ips       = (known after apply)
  + security_groups              = (known after apply)
  + source_dest_check           = true
  + spot_instance_request_id    = (known after apply)
```

```

+ subnet_id                = "subnet-0caf4afde5ceb4dce"
+ tags                      = {
  + "Name" = "terraform-ec2"
}
+ tags_all                  = {
  + "Name" = "terraform-ec2"
}
+ tenancy                   = (known after apply)
+ user_data_base64          = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids    = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ primary_network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)

+ secondary_network_interface (known after apply)
}

# aws_security_group.ec2_sg will be created
+ resource "aws_security_group" "ec2_sg" {
  + arn                = (known after apply)
  + description         = "Allow SSH"
  + egress              = [
    + {
      + cidr_blocks      = [
        + "0.0.0.0/0",
      ]
      + from_port         = 0
      + ipv6_cidr_blocks = []
      + prefix_list_ids   = []
      + protocol          = "-1"
      + security_groups   = []
      + self              = false
      + to_port           = 0
    }
  ]
}

```

```

        # (1 unchanged attribute hidden)
      },
    ],
    + id = (known after apply)
    + ingress = [
      + {
        + cidr_blocks = [
          + "0.0.0.0/0",
        ]
        + description = "SSH"
        + from_port = 22
        + ipv6_cidr_blocks = []
        + prefix_list_ids = []
        + protocol = "tcp"
        + security_groups = []
        + self = false
        + to_port = 22
      },
    ]
    + name = "ec2-sg"
    + name_prefix = (known after apply)
    + owner_id = (known after apply)
    + region = "ap-south-2"
    + revoke_rules_on_delete = false
    + tags = {
      + "Name" = "ec2-sg"
    }
    + tags_all = {
      + "Name" = "ec2-sg"
    }
    + vpc_id = "vpc-0ed3dccd5d799fbb9"
  }
}

```

Plan: 2 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```

[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform apply
aws_vpc.main: Refreshing state... [id=vpc-0ed3dccd5d799fbb9]
aws_subnet.public: Refreshing state... [id=subnet-0caf4afde5ceb4dce]
aws_internet_gateway.igw: Refreshing state... [id=igw-0ce5164dc047e0a65]
aws_route_table.public: Refreshing state... [id=rtb-0e058852b8dad8c2d]
aws_route_table_association.public: Refreshing state...
[id=rtbassoc-04a813e5b7f3a67bf]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```

# aws_instance.demo will be created
+ resource "aws_instance" "demo" {
  + ami                                = "ami-04a64102b8022e4f3"
  + arn                                = (known after apply)
  + associate_public_ip_address       = true
  + availability_zone                  = (known after apply)
  + disable_api_stop                   = (known after apply)
  + disable_api_termination            = (known after apply)
  + ebs_optimized                      = (known after apply)
  + enable_primary_ipv6                = (known after apply)
  + force_destroy                      = false
  + get_password_data                  = false
  + host_id                           = (known after apply)
  + host_resource_group_arn            = (known after apply)
  + iam_instance_profile                = (known after apply)
  + id                                 = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle                 = (known after apply)
  + instance_state                     = (known after apply)
  + instance_type                      = "t3.micro"
  + ipv6_address_count                  = (known after apply)
  + ipv6_addresses                     = (known after apply)
  + key_name                           = (known after apply)
  + monitoring                         = (known after apply)
  + outpost_arn                        = (known after apply)
  + password_data                      = (known after apply)
  + placement_group                    = (known after apply)
  + placement_group_id                 = (known after apply)
  + placement_partition_number          = (known after apply)
  + primary_network_interface_id        = (known after apply)
  + private_dns                        = (known after apply)
  + private_ip                         = (known after apply)
  + public_dns                         = (known after apply)
  + public_ip                          = (known after apply)
  + region                             = "ap-south-2"
  + secondary_private_ips               = (known after apply)
  + security_groups                    = (known after apply)
  + source_dest_check                  = true
  + spot_instance_request_id           = (known after apply)
  + subnet_id                          = "subnet-0caf4afde5ceb4dce"
  + tags                               = {
    + "Name" = "terraform-ec2"
  }
  + tags_all                           = {
    + "Name" = "terraform-ec2"
  }
  + tenancy                           = (known after apply)
  + user_data_base64                  = (known after apply)
  + user_data_replace_on_change       = false
  + vpc_security_group_ids            = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

```

```

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ primary_network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)

+ secondary_network_interface (known after apply)
}

# aws_security_group.ec2_sg will be created
+ resource "aws_security_group" "ec2_sg" {
+   arn                        = (known after apply)
+   description                = "Allow SSH"
+   egress                     = [
+     {
+       + cidr_blocks          = [
+         + "0.0.0.0/0",
+       ]
+       + from_port             = 0
+       + ipv6_cidr_blocks      = []
+       + prefix_list_ids       = []
+       + protocol              = "-1"
+       + security_groups       = []
+       + self                  = false
+       + to_port               = 0
+       # (1 unchanged attribute hidden)
+     },
+   ]
+   id                        = (known after apply)
+   ingress                   = [
+     {
+       + cidr_blocks          = [
+         + "0.0.0.0/0",
+       ]
+       + description          = "SSH"
+       + from_port            = 22
+       + ipv6_cidr_blocks      = []
+       + prefix_list_ids       = []
+       + protocol              = "tcp"
+       + security_groups       = []
+       + self                  = false

```

```

        + to_port          = 22
      },
    ]
  + name                  = "ec2-sg"
  + name_prefix           = (known after apply)
  + owner_id              = (known after apply)
  + region                = "ap-south-2"
  + revoke_rules_on_delete = false
  + tags                  = {
    + "Name" = "ec2-sg"
  }
  + tags_all              = {
    + "Name" = "ec2-sg"
  }
  + vpc_id                = "vpc-0ed3dccd5d799fbb9"
}

```

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_security_group.ec2_sg: Creating...

aws_security_group.ec2_sg: Creation complete after 2s [id=sg-010fa4c1c81ca32a2]

aws_instance.demo: Creating...

aws_instance.demo: Still creating... [00m10s elapsed]

aws_instance.demo: Creation complete after 12s [id=i-0ef4f6ea0e6abc50b]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

[terraform-user@ip-172-31-5-142 terraform-project]\$

[terraform-user@ip-172-31-5-142 terraform-project]\$ vi ec2.tf

[terraform-user@ip-172-31-5-142 terraform-project]\$ terraform plan

aws_vpc.main: Refreshing state... [id=vpc-0ed3dccd5d799fbb9]

aws_internet_gateway.igw: Refreshing state... [id=igw-0ce5164dc047e0a65]

aws_subnet.public: Refreshing state... [id=subnet-0caf4afde5ceb4dce]

aws_security_group.ec2_sg: Refreshing state... [id=sg-010fa4c1c81ca32a2]

aws_route_table.public: Refreshing state... [id=rtb-0e058852b8dad8c2d]

aws_instance.demo: Refreshing state... [id=i-0ef4f6ea0e6abc50b]

aws_route_table_association.public: Refreshing state...
[id=rtbassoc-04a813e5b7f3a67bf]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

-/+ destroy and then create replacement

Terraform will perform the following actions:

```

# aws_instance.demo must be replaced
-/+ resource "aws_instance" "demo" {
    ~ arn          =
    "arn:aws:ec2:ap-south-2:183192605590:instance/i-0ef4f6ea0e6abc50b" -> (known after
    apply)

```

```

~ associate_public_ip_address      = true -> false # forces replacement
~ availability_zone                 = "ap-south-2a" -> (known after apply)
~ disable_api_stop                 = false -> (known after apply)
~ disable_api_termination          = false -> (known after apply)
~ ebs_optimized                    = false -> (known after apply)
+ enable_primary_ipv6              = (known after apply)
- hibernation                      = false -> null
+ host_id                         = (known after apply)
+ host_resource_group_arn          = (known after apply)
+ iam_instance_profile             = (known after apply)
~ id                               = "i-0ef4f6ea0e6abc50b" -> (known after
apply)
~ instance_initiated_shutdown_behavior = "stop" -> (known after apply)
+ instance_lifecycle               = (known after apply)
~ instance_state                   = "running" -> (known after apply)
~ ipv6_address_count               = 0 -> (known after apply)
~ ipv6_addresses                   = [] -> (known after apply)
+ key_name                        = (known after apply)
~ monitoring                      = false -> (known after apply)
+ outpost_arn                     = (known after apply)
+ password_data                   = (known after apply)
+ placement_group                  = (known after apply)
+ placement_group_id              = (known after apply)
~ placement_partition_number       = 0 -> (known after apply)
~ primary_network_interface_id     = "eni-0c853203f8520196c" -> (known after
apply)
~ private_dns                      =
"ip-10-0-1-177.ap-south-2.compute.internal" -> (known after apply)
~ private_ip                      = "10.0.1.177" -> (known after apply)
+ public_dns                      = (known after apply)
~ public_ip                       = "40.192.2.35" -> (known after apply)
~ secondary_private_ips            = [] -> (known after apply)
~ security_groups                  = [] -> (known after apply)
+ spot_instance_request_id         = (known after apply)
tags                               = {
  "Name" = "terraform-ec2"
}
~ tenancy                         = "default" -> (known after apply)
+ user_data_base64                = (known after apply)
# (10 unchanged attributes hidden)

~ capacity_reservation_specification (known after apply)
- capacity_reservation_specification {
  - capacity_reservation_preference = "open" -> null
}

~ cpu_options (known after apply)
- cpu_options {
  - core_count          = 1 -> null
  - threads_per_core    = 2 -> null
  # (2 unchanged attributes hidden)
}

- credit_specification {
  - cpu_credits = "unlimited" -> null

```

```

    }

~ ebs_block_device (known after apply)

~ enclave_options (known after apply)
- enclave_options {
  - enabled = false -> null
}

~ ephemeral_block_device (known after apply)

~ instance_market_options (known after apply)

~ maintenance_options (known after apply)
- maintenance_options {
  - auto_recovery = "default" -> null
}

~ metadata_options (known after apply)
- metadata_options {
  - http_endpoint           = "enabled" -> null
  - http_protocol_ipv6      = "disabled" -> null
  - http_put_response_hop_limit = 2 -> null
  - http_tokens             = "required" -> null
  - instance_metadata_tags   = "disabled" -> null
}

~ network_interface (known after apply)

~ primary_network_interface (known after apply)
- primary_network_interface {
  - delete_on_termination = true -> null
  - network_interface_id  = "eni-0c853203f8520196c" -> null
}

~ private_dns_name_options (known after apply)
- private_dns_name_options {
  - enable_resource_name_dns_a_record      = false -> null
  - enable_resource_name_dns_aaaa_record   = false -> null
  - hostname_type                          = "ip-name" -> null
}

~ root_block_device (known after apply)
- root_block_device {
  - delete_on_termination = true -> null
  - device_name           = "/dev/xvda" -> null
  - encrypted             = false -> null
  - iops                  = 3000 -> null
  - tags                  = {} -> null
  - tags_all              = {} -> null
  - throughput            = 125 -> null
  - volume_id             = "vol-0c83bdda443be6d17" -> null
  - volume_size           = 8 -> null
  - volume_type           = "gp3" -> null
  # (1 unchanged attribute hidden)

```

```

    }

    ~ secondary_network_interface (known after apply)
}

```

Plan: 1 to add, 0 to change, 1 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```

[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform apply
aws_vpc.main: Refreshing state... [id=vpc-0ed3dccd5d799fbb9]
aws_internet_gateway.igw: Refreshing state... [id=igw-0ce5164dc047e0a65]
aws_subnet.public: Refreshing state... [id=subnet-0caf4afde5ceb4dce]
aws_security_group.ec2_sg: Refreshing state... [id=sg-010fa4c1c81ca32a2]
aws_route_table.public: Refreshing state... [id=rtb-0e058852b8dad8c2d]
aws_route_table_association.public: Refreshing state...
[id=rtbassoc-04a813e5b7f3a67bf]
aws_instance.demo: Refreshing state... [id=i-0ef4f6ea0e6abc50b]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

-/+ destroy and then create replacement

Terraform will perform the following actions:

```

# aws_instance.demo must be replaced
-/+ resource "aws_instance" "demo" {
    ~ arn =
    "arn:aws:ec2:ap-south-2:183192605590:instance/i-0ef4f6ea0e6abc50b" -> (known after
    apply)
    ~ associate_public_ip_address = true -> false # forces replacement
    ~ availability_zone            = "ap-south-2a" -> (known after apply)
    ~ disable_api_stop             = false -> (known after apply)
    ~ disable_api_termination     = false -> (known after apply)
    ~ ebs_optimized               = false -> (known after apply)
    + enable_primary_ipv6         = (known after apply)
    - hibernation                 = false -> null
    + host_id                     = (known after apply)
    + host_resource_group_arn     = (known after apply)
    + iam_instance_profile        = (known after apply)
    ~ id                          = "i-0ef4f6ea0e6abc50b" -> (known after
    apply)
    ~ instance_initiated_shutdown_behavior = "stop" -> (known after apply)
    + instance_lifecycle          = (known after apply)
    ~ instance_state              = "running" -> (known after apply)
    ~ ipv6_address_count          = 0 -> (known after apply)
    ~ ipv6_addresses              = [] -> (known after apply)
    + key_name                    = (known after apply)
    ~ monitoring                   = false -> (known after apply)
    + outpost_arn                 = (known after apply)
    + password_data               = (known after apply)

```

```

+ placement_group                = (known after apply)
+ placement_group_id             = (known after apply)
~ placement_partition_number     = 0 -> (known after apply)
~ primary_network_interface_id   = "eni-0c853203f8520196c" -> (known after
apply)
~ private_dns                    =
"ip-10-0-1-177.ap-south-2.compute.internal" -> (known after apply)
~ private_ip                    = "10.0.1.177" -> (known after apply)
+ public_dns                    = (known after apply)
~ public_ip                     = "40.192.2.35" -> (known after apply)
~ secondary_private_ips         = [] -> (known after apply)
~ security_groups               = [] -> (known after apply)
+ spot_instance_request_id      = (known after apply)
  tags                          = {
    "Name" = "terraform-ec2"
  }
~ tenancy                      = "default" -> (known after apply)
+ user_data_base64             = (known after apply)
  # (10 unchanged attributes hidden)

~ capacity_reservation_specification (known after apply)
- capacity_reservation_specification {
  - capacity_reservation_preference = "open" -> null
}

~ cpu_options (known after apply)
- cpu_options {
  - core_count          = 1 -> null
  - threads_per_core    = 2 -> null
  # (2 unchanged attributes hidden)
}

- credit_specification {
  - cpu_credits = "unlimited" -> null
}

~ ebs_block_device (known after apply)

~ enclave_options (known after apply)
- enclave_options {
  - enabled = false -> null
}

~ ephemeral_block_device (known after apply)

~ instance_market_options (known after apply)

~ maintenance_options (known after apply)
- maintenance_options {
  - auto_recovery = "default" -> null
}

~ metadata_options (known after apply)
- metadata_options {
  - http_endpoint          = "enabled" -> null

```

```

- http_protocol_ipv6          = "disabled" -> null
- http_put_response_hop_limit = 2 -> null
- http_tokens                 = "required" -> null
- instance_metadata_tags      = "disabled" -> null
}

~ network_interface (known after apply)

~ primary_network_interface (known after apply)
- primary_network_interface {
  - delete_on_termination = true -> null
  - network_interface_id  = "eni-0c853203f8520196c" -> null
}

~ private_dns_name_options (known after apply)
- private_dns_name_options {
  - enable_resource_name_dns_a_record    = false -> null
  - enable_resource_name_dns_aaaa_record = false -> null
  - hostname_type                       = "ip-name" -> null
}

~ root_block_device (known after apply)
- root_block_device {
  - delete_on_termination = true -> null
  - device_name           = "/dev/xvda" -> null
  - encrypted             = false -> null
  - iops                  = 3000 -> null
  - tags                  = {} -> null
  - tags_all              = {} -> null
  - throughput            = 125 -> null
  - volume_id             = "vol-0c83bdda443be6d17" -> null
  - volume_size           = 8 -> null
  - volume_type           = "gp3" -> null
  # (1 unchanged attribute hidden)
}

~ secondary_network_interface (known after apply)
}

```

Plan: 1 to add, 0 to change, 1 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

```

aws_instance.demo: Destroying... [id=i-0ef4f6ea0e6abc50b]
aws_instance.demo: Still destroying... [id=i-0ef4f6ea0e6abc50b, 00m10s elapsed]
aws_instance.demo: Still destroying... [id=i-0ef4f6ea0e6abc50b, 00m20s elapsed]
aws_instance.demo: Destruction complete after 30s
aws_instance.demo: Creating...
aws_instance.demo: Still creating... [00m10s elapsed]
aws_instance.demo: Creation complete after 12s [id=i-0529f98074e14d755]

```

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.

```
[terraform-user@ip-172-31-5-142 terraform-project]$ Read from remote host  
ec2-16-112-129-181.ap-south-2.compute.amazonaws.com: Connection reset by peer  
Connection to ec2-16-112-129-181.ap-south-2.compute.amazonaws.com closed.  
client_loop: send disconnect: Connection reset by peer
```

X

📅 2026-06-20 ▶ 🕒 18:02.33 ▶ 📁 /drives/c/Users/USER/Downloads ▶

```
[terraform-user@ip-172-31-5-142 terraform-project]$ vi provider.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ vi vpc.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ vi vpc.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ vi vpc.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform init
Initializing provider plugins found in the configuration...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.51.0
```

Initializing the backend...

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform validate
Success! The configuration is valid.
```

```
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform plan
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```
# aws_internet_gateway.igw will be created
+ resource "aws_internet_gateway" "igw" {
  + arn      = (known after apply)
  + id       = (known after apply)
  + owner_id = (known after apply)
  + region   = "ap-south-2"
  + tags     = {
    + "Name" = "terraform-igw"
  }
  + tags_all = {
    + "Name" = "terraform-igw"
  }
  + vpc_id   = (known after apply)
}

# aws_route_table.public will be created
+ resource "aws_route_table" "public" {
  + arn              = (known after apply)
  + id               = (known after apply)
  + owner_id         = (known after apply)
  + propagating_vgws = (known after apply)
  + region           = "ap-south-2"
```

```

+ route          = [
+   {
+     + cidr_block          = "0.0.0.0/0"
+     + gateway_id          = (known after apply)
+     # (12 unchanged attributes hidden)
+   },
+ ]
+ tags            = {
+   + "Name" = "public-rt"
+ }
+ tags_all        = {
+   + "Name" = "public-rt"
+ }
+ vpc_id          = (known after apply)
}

# aws_route_table_association.public will be created
+ resource "aws_route_table_association" "public" {
+   id            = (known after apply)
+   region        = "ap-south-2"
+   route_table_id = (known after apply)
+   subnet_id     = (known after apply)
+ }

# aws_subnet.public will be created
+ resource "aws_subnet" "public" {
+   arn                                = (known after apply)
+   assign_ipv6_address_on_creation    = false
+   availability_zone                  = "ap-south-2a"
+   availability_zone_id                = (known after apply)
+   cidr_block                         = "10.0.1.0/24"
+   enable_dns64                       = false
+   enable_resource_name_dns_a_record_on_launch = false
+   enable_resource_name_dns_aaaa_record_on_launch = false
+   id                                = (known after apply)
+   ipv6_cidr_block                    = (known after apply)
+   ipv6_cidr_block_association_id     = (known after apply)
+   ipv6_native                        = false
+   map_public_ip_on_launch            = true
+   owner_id                          = (known after apply)
+   private_dns_hostname_type_on_launch = (known after apply)
+   region                            = "ap-south-2"
+   tags                               = {
+     + "Name" = "public-subnet"
+   }
+   tags_all                          = {
+     + "Name" = "public-subnet"
+   }
+   vpc_id                            = (known after apply)
+ }

# aws_vpc.main will be created
+ resource "aws_vpc" "main" {
+   arn          = (known after apply)
+   cidr_block    = "10.0.0.0/16"

```

```

+ default_network_acl_id          = (known after apply)
+ default_route_table_id         = (known after apply)
+ default_security_group_id      = (known after apply)
+ dhcp_options_id                = (known after apply)
+ enable_dns_hostnames           = (known after apply)
+ enable_dns_support             = true
+ enable_network_address_usage_metrics = (known after apply)
+ id                             = (known after apply)
+ instance_tenancy               = "default"
+ ipv6_association_id            = (known after apply)
+ ipv6_cidr_block                = (known after apply)
+ ipv6_cidr_block_network_border_group = (known after apply)
+ main_route_table_id           = (known after apply)
+ owner_id                      = (known after apply)
+ region                         = "ap-south-2"
+ tags                           = {
    + "Name" = "terraform-vpc"
  }
+ tags_all                       = {
    + "Name" = "terraform-vpc"
  }
}

```

Plan: 5 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform apply
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

aws_internet_gateway.igw will be created

```

+ resource "aws_internet_gateway" "igw" {
  + arn      = (known after apply)
  + id       = (known after apply)
  + owner_id = (known after apply)
  + region   = "ap-south-2"
  + tags     = {
    + "Name" = "terraform-igw"
  }
  + tags_all = {
    + "Name" = "terraform-igw"
  }
  + vpc_id   = (known after apply)
}

```

aws_route_table.public will be created

```

+ resource "aws_route_table" "public" {

```

```

+ arn                = (known after apply)
+ id                 = (known after apply)
+ owner_id           = (known after apply)
+ propagating_vgws   = (known after apply)
+ region             = "ap-south-2"
+ route              = [
  + {
    + cidr_block       = "0.0.0.0/0"
    + gateway_id       = (known after apply)
    # (12 unchanged attributes hidden)
  },
]
+ tags               = {
  + "Name" = "public-rt"
}
+ tags_all           = {
  + "Name" = "public-rt"
}
+ vpc_id             = (known after apply)
}

# aws_route_table_association.public will be created
+ resource "aws_route_table_association" "public" {
  + id                = (known after apply)
  + region            = "ap-south-2"
  + route_table_id    = (known after apply)
  + subnet_id         = (known after apply)
}

# aws_subnet.public will be created
+ resource "aws_subnet" "public" {
  + arn                = (known after apply)
  + assign_ipv6_address_on_creation = false
  + availability_zone   = "ap-south-2a"
  + availability_zone_id = (known after apply)
  + cidr_block          = "10.0.1.0/24"
  + enable_dns64         = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                 = (known after apply)
  + ipv6_cidr_block     = (known after apply)
  + ipv6_cidr_block_association_id = (known after apply)
  + ipv6_native         = false
  + map_public_ip_on_launch = true
  + owner_id           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region             = "ap-south-2"
  + tags               = {
    + "Name" = "public-subnet"
  }
  + tags_all           = {
    + "Name" = "public-subnet"
  }
  + vpc_id             = (known after apply)
}

```

```
# aws_vpc.main will be created
+ resource "aws_vpc" "main" {
  + arn                                = (known after apply)
  + cidr_block                        = "10.0.0.0/16"
  + default_network_acl_id           = (known after apply)
  + default_route_table_id           = (known after apply)
  + default_security_group_id        = (known after apply)
  + dhcp_options_id                  = (known after apply)
  + enable_dns_hostnames              = (known after apply)
  + enable_dns_support                = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                                = (known after apply)
  + instance_tenancy                  = "default"
  + ipv6_association_id               = (known after apply)
  + ipv6_cidr_block                   = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id               = (known after apply)
  + owner_id                          = (known after apply)
  + region                            = "ap-south-2"
  + tags                              = {
    + "Name" = "terraform-vpc"
  }
  + tags_all                          = {
    + "Name" = "terraform-vpc"
  }
}
```

Plan: 5 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

aws_vpc.main: Creating...

aws_vpc.main: Creation complete after 1s [id=vpc-0ed3dccc5d799fbb9]

aws_internet_gateway.igw: Creating...

aws_subnet.public: Creating...

aws_internet_gateway.igw: Creation complete after 0s [id=igw-0ce5164dc047e0a65]

aws_route_table.public: Creating...

aws_route_table.public: Creation complete after 1s [id=rtb-0e058852b8dad8c2d]

aws_subnet.public: Still creating... [00m10s elapsed]

aws_subnet.public: Creation complete after 11s [id=subnet-0caf4afde5ceb4dce]

aws_route_table_association.public: Creating...

aws_route_table_association.public: Creation complete after 0s

[id=rtbassoc-04a813e5b7f3a67bf]

Apply complete! Resources: 5 added, 0 changed, 0 destroyed.

[terraform-user@ip-172-31-5-142 terraform-project]\$

[terraform-user@ip-172-31-5-142 terraform-project]\$ vi security.tf

[terraform-user@ip-172-31-5-142 terraform-project]\$ vi ec2.tf

[terraform-user@ip-172-31-5-142 terraform-project]\$ terraform fmt

[terraform-user@ip-172-31-5-142 terraform-project]\$ terraform validate

Success! The configuration is valid.

```
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform plan
aws_vpc.main: Refreshing state... [id=vpc-0ed3dccd5d799fbb9]
aws_subnet.public: Refreshing state... [id=subnet-0caf4afde5ceb4dce]
aws_internet_gateway.igw: Refreshing state... [id=igw-0ce5164dc047e0a65]
aws_route_table.public: Refreshing state... [id=rtb-0e058852b8dad8c2d]
aws_route_table_association.public: Refreshing state...
[id=rtbassoc-04a813e5b7f3a67bf]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_instance.demo will be created
+ resource "aws_instance" "demo" {
  + ami                      = "ami-04a64102b8022e4f3"
  + arn                     = (known after apply)
  + associate_public_ip_address = true
  + availability_zone        = (known after apply)
  + disable_api_stop         = (known after apply)
  + disable_api_termination  = (known after apply)
  + ebs_optimized            = (known after apply)
  + enable_primary_ipv6      = (known after apply)
  + force_destroy            = false
  + get_password_data        = false
  + host_id                  = (known after apply)
  + host_resource_group_arn  = (known after apply)
  + iam_instance_profile     = (known after apply)
  + id                      = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle       = (known after apply)
  + instance_state           = (known after apply)
  + instance_type            = "t3.micro"
  + ipv6_address_count        = (known after apply)
  + ipv6_addresses           = (known after apply)
  + key_name                  = (known after apply)
  + monitoring                = (known after apply)
  + outpost_arn              = (known after apply)
  + password_data            = (known after apply)
  + placement_group          = (known after apply)
  + placement_group_id       = (known after apply)
  + placement_partition_number = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns               = (known after apply)
  + private_ip                = (known after apply)
  + public_dns                = (known after apply)
  + public_ip                 = (known after apply)
  + region                    = "ap-south-2"
  + secondary_private_ips     = (known after apply)
  + security_groups           = (known after apply)
  + source_dest_check         = true
  + spot_instance_request_id  = (known after apply)
```

```

+ subnet_id                = "subnet-0caf4afde5ceb4dce"
+ tags                      = {
  + "Name" = "terraform-ec2"
}
+ tags_all                  = {
  + "Name" = "terraform-ec2"
}
+ tenancy                   = (known after apply)
+ user_data_base64          = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids    = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ primary_network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)

+ secondary_network_interface (known after apply)
}

# aws_security_group.ec2_sg will be created
+ resource "aws_security_group" "ec2_sg" {
  + arn                = (known after apply)
  + description         = "Allow SSH"
  + egress              = [
    + {
      + cidr_blocks      = [
        + "0.0.0.0/0",
      ]
      + from_port         = 0
      + ipv6_cidr_blocks = []
      + prefix_list_ids   = []
      + protocol          = "-1"
      + security_groups   = []
      + self              = false
      + to_port           = 0
    }
  ]
}

```

```

        # (1 unchanged attribute hidden)
    },
]
+ id = (known after apply)
+ ingress = [
  + {
    + cidr_blocks = [
      + "0.0.0.0/0",
    ]
    + description = "SSH"
    + from_port = 22
    + ipv6_cidr_blocks = []
    + prefix_list_ids = []
    + protocol = "tcp"
    + security_groups = []
    + self = false
    + to_port = 22
  },
]
+ name = "ec2-sg"
+ name_prefix = (known after apply)
+ owner_id = (known after apply)
+ region = "ap-south-2"
+ revoke_rules_on_delete = false
+ tags = {
  + "Name" = "ec2-sg"
}
+ tags_all = {
  + "Name" = "ec2-sg"
}
+ vpc_id = "vpc-0ed3dccd5d799fbb9"
}

```

Plan: 2 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```

[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform apply
aws_vpc.main: Refreshing state... [id=vpc-0ed3dccd5d799fbb9]
aws_subnet.public: Refreshing state... [id=subnet-0caf4afde5ceb4dce]
aws_internet_gateway.igw: Refreshing state... [id=igw-0ce5164dc047e0a65]
aws_route_table.public: Refreshing state... [id=rtb-0e058852b8dad8c2d]
aws_route_table_association.public: Refreshing state...
[id=rtbassoc-04a813e5b7f3a67bf]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```

# aws_instance.demo will be created
+ resource "aws_instance" "demo" {
  + ami                                = "ami-04a64102b8022e4f3"
  + arn                                = (known after apply)
  + associate_public_ip_address       = true
  + availability_zone                  = (known after apply)
  + disable_api_stop                   = (known after apply)
  + disable_api_termination            = (known after apply)
  + ebs_optimized                      = (known after apply)
  + enable_primary_ipv6                = (known after apply)
  + force_destroy                      = false
  + get_password_data                  = false
  + host_id                            = (known after apply)
  + host_resource_group_arn            = (known after apply)
  + iam_instance_profile                = (known after apply)
  + id                                 = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle                 = (known after apply)
  + instance_state                     = (known after apply)
  + instance_type                      = "t3.micro"
  + ipv6_address_count                 = (known after apply)
  + ipv6_addresses                     = (known after apply)
  + key_name                           = (known after apply)
  + monitoring                         = (known after apply)
  + outpost_arn                       = (known after apply)
  + password_data                      = (known after apply)
  + placement_group                   = (known after apply)
  + placement_group_id                 = (known after apply)
  + placement_partition_number         = (known after apply)
  + primary_network_interface_id       = (known after apply)
  + private_dns                        = (known after apply)
  + private_ip                         = (known after apply)
  + public_dns                         = (known after apply)
  + public_ip                          = (known after apply)
  + region                             = "ap-south-2"
  + secondary_private_ips               = (known after apply)
  + security_groups                    = (known after apply)
  + source_dest_check                  = true
  + spot_instance_request_id           = (known after apply)
  + subnet_id                          = "subnet-0caf4afde5ceb4dce"
  + tags                               = {
    + "Name" = "terraform-ec2"
  }
  + tags_all                           = {
    + "Name" = "terraform-ec2"
  }
  + tenancy                            = (known after apply)
  + user_data_base64                   = (known after apply)
  + user_data_replace_on_change        = false
  + vpc_security_group_ids             = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

```

```

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ primary_network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)

+ secondary_network_interface (known after apply)
}

# aws_security_group.ec2_sg will be created
+ resource "aws_security_group" "ec2_sg" {
  + arn                = (known after apply)
  + description        = "Allow SSH"
  + egress              = [
    + {
      + cidr_blocks      = [
        + "0.0.0.0/0",
      ]
      + from_port         = 0
      + ipv6_cidr_blocks = []
      + prefix_list_ids   = []
      + protocol          = "-1"
      + security_groups   = []
      + self              = false
      + to_port           = 0
      # (1 unchanged attribute hidden)
    },
  ]
  + id                  = (known after apply)
  + ingress              = [
    + {
      + cidr_blocks      = [
        + "0.0.0.0/0",
      ]
      + description       = "SSH"
      + from_port         = 22
      + ipv6_cidr_blocks = []
      + prefix_list_ids   = []
      + protocol          = "tcp"
      + security_groups   = []
      + self              = false
    }
  ]
}

```

```

        + to_port          = 22
      },
    ]
+ name                    = "ec2-sg"
+ name_prefix            = (known after apply)
+ owner_id               = (known after apply)
+ region                 = "ap-south-2"
+ revoke_rules_on_delete = false
+ tags                   = {
    + "Name" = "ec2-sg"
  }
+ tags_all               = {
    + "Name" = "ec2-sg"
  }
+ vpc_id                 = "vpc-0ed3dccd5d799fbb9"
}

```

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_security_group.ec2_sg: Creating...

aws_security_group.ec2_sg: Creation complete after 2s [id=sg-010fa4c1c81ca32a2]

aws_instance.demo: Creating...

aws_instance.demo: Still creating... [00m10s elapsed]

aws_instance.demo: Creation complete after 12s [id=i-0ef4f6ea0e6abc50b]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

[terraform-user@ip-172-31-5-142 terraform-project]\$

[terraform-user@ip-172-31-5-142 terraform-project]\$ vi ec2.tf

[terraform-user@ip-172-31-5-142 terraform-project]\$ terraform plan

aws_vpc.main: Refreshing state... [id=vpc-0ed3dccd5d799fbb9]

aws_internet_gateway.igw: Refreshing state... [id=igw-0ce5164dc047e0a65]

aws_subnet.public: Refreshing state... [id=subnet-0caf4afde5ceb4dce]

aws_security_group.ec2_sg: Refreshing state... [id=sg-010fa4c1c81ca32a2]

aws_route_table.public: Refreshing state... [id=rtb-0e058852b8dad8c2d]

aws_instance.demo: Refreshing state... [id=i-0ef4f6ea0e6abc50b]

aws_route_table_association.public: Refreshing state...

[id=rtbassoc-04a813e5b7f3a67bf]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

-/+ destroy and then create replacement

Terraform will perform the following actions:

aws_instance.demo must be replaced

-/+ resource "aws_instance" "demo" {

~ arn

=

"arn:aws:ec2:ap-south-2:183192605590:instance/i-0ef4f6ea0e6abc50b" -> (known after apply)

```

~ associate_public_ip_address      = true -> false # forces replacement
~ availability_zone                 = "ap-south-2a" -> (known after apply)
~ disable_api_stop                  = false -> (known after apply)
~ disable_api_termination           = false -> (known after apply)
~ ebs_optimized                     = false -> (known after apply)
+ enable_primary_ipv6               = (known after apply)
- hibernation                       = false -> null
+ host_id                           = (known after apply)
+ host_resource_group_arn           = (known after apply)
+ iam_instance_profile              = (known after apply)
~ id                                = "i-0ef4f6ea0e6abc50b" -> (known after
apply)
~ instance_initiated_shutdown_behavior = "stop" -> (known after apply)
+ instance_lifecycle                 = (known after apply)
~ instance_state                     = "running" -> (known after apply)
~ ipv6_address_count                 = 0 -> (known after apply)
~ ipv6_addresses                     = [] -> (known after apply)
+ key_name                           = (known after apply)
~ monitoring                         = false -> (known after apply)
+ outpost_arn                       = (known after apply)
+ password_data                      = (known after apply)
+ placement_group                    = (known after apply)
+ placement_group_id                 = (known after apply)
~ placement_partition_number         = 0 -> (known after apply)
~ primary_network_interface_id       = "eni-0c853203f8520196c" -> (known after
apply)
~ private_dns                        =
"ip-10-0-1-177.ap-south-2.compute.internal" -> (known after apply)
~ private_ip                         = "10.0.1.177" -> (known after apply)
+ public_dns                         = (known after apply)
~ public_ip                          = "40.192.2.35" -> (known after apply)
~ secondary_private_ips               = [] -> (known after apply)
~ security_groups                     = [] -> (known after apply)
+ spot_instance_request_id            = (known after apply)
tags                                 = {
  "Name" = "terraform-ec2"
}
~ tenancy                             = "default" -> (known after apply)
+ user_data_base64                     = (known after apply)
# (10 unchanged attributes hidden)

~ capacity_reservation_specification (known after apply)
- capacity_reservation_specification {
  - capacity_reservation_preference = "open" -> null
}

~ cpu_options (known after apply)
- cpu_options {
  - core_count          = 1 -> null
  - threads_per_core    = 2 -> null
  # (2 unchanged attributes hidden)
}

- credit_specification {
  - cpu_credits = "unlimited" -> null

```

```

}

~ ebs_block_device (known after apply)

~ enclave_options (known after apply)
- enclave_options {
  - enabled = false -> null
}

~ ephemeral_block_device (known after apply)

~ instance_market_options (known after apply)

~ maintenance_options (known after apply)
- maintenance_options {
  - auto_recovery = "default" -> null
}

~ metadata_options (known after apply)
- metadata_options {
  - http_endpoint           = "enabled" -> null
  - http_protocol_ipv6      = "disabled" -> null
  - http_put_response_hop_limit = 2 -> null
  - http_tokens             = "required" -> null
  - instance_metadata_tags   = "disabled" -> null
}

~ network_interface (known after apply)

~ primary_network_interface (known after apply)
- primary_network_interface {
  - delete_on_termination = true -> null
  - network_interface_id  = "eni-0c853203f8520196c" -> null
}

~ private_dns_name_options (known after apply)
- private_dns_name_options {
  - enable_resource_name_dns_a_record    = false -> null
  - enable_resource_name_dns_aaaa_record = false -> null
  - hostname_type                        = "ip-name" -> null
}

~ root_block_device (known after apply)
- root_block_device {
  - delete_on_termination = true -> null
  - device_name           = "/dev/xvda" -> null
  - encrypted             = false -> null
  - iops                  = 3000 -> null
  - tags                  = {} -> null
  - tags_all              = {} -> null
  - throughput            = 125 -> null
  - volume_id             = "vol-0c83bdda443be6d17" -> null
  - volume_size           = 8 -> null
  - volume_type           = "gp3" -> null
  # (1 unchanged attribute hidden)

```

```

    }

    ~ secondary_network_interface (known after apply)
}

```

Plan: 1 to add, 0 to change, 1 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```

[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform apply
aws_vpc.main: Refreshing state... [id=vpc-0ed3dccd5d799fbb9]
aws_internet_gateway.igw: Refreshing state... [id=igw-0ce5164dc047e0a65]
aws_subnet.public: Refreshing state... [id=subnet-0caf4afde5ceb4dce]
aws_security_group.ec2_sg: Refreshing state... [id=sg-010fa4c1c81ca32a2]
aws_route_table.public: Refreshing state... [id=rtb-0e058852b8dad8c2d]
aws_route_table_association.public: Refreshing state...
[id=rtbassoc-04a813e5b7f3a67bf]
aws_instance.demo: Refreshing state... [id=i-0ef4f6ea0e6abc50b]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

-/+ destroy and then create replacement

Terraform will perform the following actions:

```

# aws_instance.demo must be replaced
-/+ resource "aws_instance" "demo" {
    ~ arn =
    "arn:aws:ec2:ap-south-2:183192605590:instance/i-0ef4f6ea0e6abc50b" -> (known after
    apply)
    ~ associate_public_ip_address = true -> false # forces replacement
    ~ availability_zone           = "ap-south-2a" -> (known after apply)
    ~ disable_api_stop           = false -> (known after apply)
    ~ disable_api_termination    = false -> (known after apply)
    ~ ebs_optimized              = false -> (known after apply)
    + enable_primary_ipv6        = (known after apply)
    - hibernation                = false -> null
    + host_id                    = (known after apply)
    + host_resource_group_arn    = (known after apply)
    + iam_instance_profile       = (known after apply)
    ~ id                         = "i-0ef4f6ea0e6abc50b" -> (known after
    apply)
    ~ instance_initiated_shutdown_behavior = "stop" -> (known after apply)
    + instance_lifecycle         = (known after apply)
    ~ instance_state             = "running" -> (known after apply)
    ~ ipv6_address_count         = 0 -> (known after apply)
    ~ ipv6_addresses            = [] -> (known after apply)
    + key_name                   = (known after apply)
    ~ monitoring                 = false -> (known after apply)
    + outpost_arn               = (known after apply)
    + password_data              = (known after apply)

```

```

+ placement_group                = (known after apply)
+ placement_group_id             = (known after apply)
~ placement_partition_number     = 0 -> (known after apply)
~ primary_network_interface_id   = "eni-0c853203f8520196c" -> (known after
apply)
~ private_dns                    =
"ip-10-0-1-177.ap-south-2.compute.internal" -> (known after apply)
~ private_ip                    = "10.0.1.177" -> (known after apply)
+ public_dns                    = (known after apply)
~ public_ip                     = "40.192.2.35" -> (known after apply)
~ secondary_private_ips         = [] -> (known after apply)
~ security_groups               = [] -> (known after apply)
+ spot_instance_request_id      = (known after apply)
tags                            = {
  "Name" = "terraform-ec2"
}
~ tenancy                      = "default" -> (known after apply)
+ user_data_base64             = (known after apply)
# (10 unchanged attributes hidden)

~ capacity_reservation_specification (known after apply)
- capacity_reservation_specification {
  - capacity_reservation_preference = "open" -> null
}

~ cpu_options (known after apply)
- cpu_options {
  - core_count          = 1 -> null
  - threads_per_core    = 2 -> null
  # (2 unchanged attributes hidden)
}

- credit_specification {
  - cpu_credits = "unlimited" -> null
}

~ ebs_block_device (known after apply)

~ enclave_options (known after apply)
- enclave_options {
  - enabled = false -> null
}

~ ephemeral_block_device (known after apply)

~ instance_market_options (known after apply)

~ maintenance_options (known after apply)
- maintenance_options {
  - auto_recovery = "default" -> null
}

~ metadata_options (known after apply)
- metadata_options {
  - http_endpoint          = "enabled" -> null

```

```

    - http_protocol_ipv6          = "disabled" -> null
    - http_put_response_hop_limit = 2 -> null
    - http_tokens                 = "required" -> null
    - instance_metadata_tags      = "disabled" -> null
  }

  ~ network_interface (known after apply)

  ~ primary_network_interface (known after apply)
  - primary_network_interface {
    - delete_on_termination = true -> null
    - network_interface_id  = "eni-0c853203f8520196c" -> null
  }

  ~ private_dns_name_options (known after apply)
  - private_dns_name_options {
    - enable_resource_name_dns_a_record    = false -> null
    - enable_resource_name_dns_aaaa_record = false -> null
    - hostname_type                       = "ip-name" -> null
  }

  ~ root_block_device (known after apply)
  - root_block_device {
    - delete_on_termination = true -> null
    - device_name           = "/dev/xvda" -> null
    - encrypted             = false -> null
    - iops                  = 3000 -> null
    - tags                  = {} -> null
    - tags_all              = {} -> null
    - throughput            = 125 -> null
    - volume_id             = "vol-0c83bdda443be6d17" -> null
    - volume_size           = 8 -> null
    - volume_type           = "gp3" -> null
    # (1 unchanged attribute hidden)
  }

  ~ secondary_network_interface (known after apply)
}

```

Plan: 1 to add, 0 to change, 1 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

```

aws_instance.demo: Destroying... [id=i-0ef4f6ea0e6abc50b]
aws_instance.demo: Still destroying... [id=i-0ef4f6ea0e6abc50b, 00m10s elapsed]
aws_instance.demo: Still destroying... [id=i-0ef4f6ea0e6abc50b, 00m20s elapsed]
aws_instance.demo: Destruction complete after 30s
aws_instance.demo: Creating...
aws_instance.demo: Still creating... [00m10s elapsed]
aws_instance.demo: Creation complete after 12s [id=i-0529f98074e14d755]

```

```
[terraform-user@ip-172-31-5-142 terraform-project]$ Read from remote host
ec2-16-112-129-181.ap-south-2.compute.amazonaws.com: Connection reset by peer
Connection to ec2-16-112-129-181.ap-south-2.compute.amazonaws.com closed.
client_loop: send disconnect: Connection reset by peer
```

📁 2026-06-20 ▶ 🕒 18:02.33 ▶ 📁 /drives/c/Users/USER/Downloads ▶ clear

```

      #_
~\   #####_           Amazon Linux 2023
~\   #####\_
~\   \###|
~\   \#/_
~    V~' '->
~~~~
~..-.-/_
~/m/'

```

```
[id=rtbassoc-04a813e5b7f3a67bf]
```

- + create

Terraform will perform the following actions:

```
# aws_s3_bucket.demo will be created
+ resource "aws_s3_bucket" "demo" {
  + acceleration_status      = (known after apply)
  + acl                      = (known after apply)
  + arn                      = (known after apply)
  + bucket                   = "sukhmay-demo-bucket-2026-001"
  + bucket_domain_name      = (known after apply)
  + bucket_namespace        = (known after apply)
  + bucket_prefix            = (known after apply)
  + bucket_region           = (known after apply)
  + bucket_regional_domain_name = (known after apply)
  + force_destroy           = false
  + hosted_zone_id          = (known after apply)
  + id                      = (known after apply)
  + object_lock_enabled     = (known after apply)
  + policy                  = (known after apply)
  + region                  = "ap-south-2"
  + request_payer           = (known after apply)
  + tags_all                = (known after apply)
  + website_domain          = (known after apply)
  + website_endpoint        = (known after apply)

  + cors_rule (known after apply)

  + grant (known after apply)

  + lifecycle_rule (known after apply)

  + logging (known after apply)

  + object_lock_configuration (known after apply)

  + replication_configuration (known after apply)

  + server_side_encryption_configuration (known after apply)

  + versioning (known after apply)

  + website (known after apply)
}

# aws_sqs_queue.demo will be created
+ resource "aws_sqs_queue" "demo" {
  + arn                      = (known after apply)
  + content_based_deduplication = false
  + deduplication_scope      = (known after apply)
  + delay_seconds            = 0
  + fifo_queue               = false
  + fifo_throughput_limit    = (known after apply)
  + id                      = (known after apply)
  + kms_data_key_reuse_period_seconds = (known after apply)
  + max_message_size         = 262144
  + message_retention_seconds = 345600
}
```

```

+ name = "demo-queue"
+ name_prefix = (known after apply)
+ policy = (known after apply)
+ receive_wait_time_seconds = 0
+ redrive_allow_policy = (known after apply)
+ redrive_policy = (known after apply)
+ region = "ap-south-2"
+ sqs_managed_sse_enabled = (known after apply)
+ tags_all = (known after apply)
+ url = (known after apply)
+ visibility_timeout_seconds = 30
}

```

Plan: 2 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```

[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform apply
aws_vpc.main: Refreshing state... [id=vpc-0ed3dccd5d799fbb9]
aws_security_group.ec2_sg: Refreshing state... [id=sg-010fa4c1c81ca32a2]
aws_internet_gateway.igw: Refreshing state... [id=igw-0ce5164dc047e0a65]
aws_subnet.public: Refreshing state... [id=subnet-0caf4afde5ceb4dce]
aws_route_table.public: Refreshing state... [id=rtb-0e058852b8dad8c2d]
aws_instance.demo: Refreshing state... [id=i-0529f98074e14d755]
aws_route_table_association.public: Refreshing state...
[id=rtbassoc-04a813e5b7f3a67bf]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```

# aws_s3_bucket.demo will be created
+ resource "aws_s3_bucket" "demo" {
+   acceleration_status = (known after apply)
+   acl                 = (known after apply)
+   arn                 = (known after apply)
+   bucket              = "sukhmay-demo-bucket-2026-001"
+   bucket_domain_name = (known after apply)
+   bucket_namespace   = (known after apply)
+   bucket_prefix       = (known after apply)
+   bucket_region       = (known after apply)
+   bucket_regional_domain_name = (known after apply)
+   force_destroy       = false
+   hosted_zone_id      = (known after apply)
+   id                  = (known after apply)
+   object_lock_enabled = (known after apply)
+   policy              = (known after apply)
+   region              = "ap-south-2"
+   request_payer       = (known after apply)

```

```

+ tags_all                = (known after apply)
+ website_domain          = (known after apply)
+ website_endpoint        = (known after apply)

+ cors_rule (known after apply)

+ grant (known after apply)

+ lifecycle_rule (known after apply)

+ logging (known after apply)

+ object_lock_configuration (known after apply)

+ replication_configuration (known after apply)

+ server_side_encryption_configuration (known after apply)

+ versioning (known after apply)

+ website (known after apply)
}

# aws_sqs_queue.demo will be created
+ resource "aws_sqs_queue" "demo" {
  + arn                = (known after apply)
  + content_based_deduplication = false
  + deduplication_scope = (known after apply)
  + delay_seconds      = 0
  + fifo_queue         = false
  + fifo_throughput_limit = (known after apply)
  + id                 = (known after apply)
  + kms_data_key_reuse_period_seconds = (known after apply)
  + max_message_size    = 262144
  + message_retention_seconds = 345600
  + name                = "demo-queue"
  + name_prefix         = (known after apply)
  + policy              = (known after apply)
  + receive_wait_time_seconds = 0
  + redrive_allow_policy = (known after apply)
  + redrive_policy      = (known after apply)
  + region              = "ap-south-2"
  + sqs_managed_sse_enabled = (known after apply)
  + tags_all            = (known after apply)
  + url                 = (known after apply)
  + visibility_timeout_seconds = 30
}

```

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

```
aws_sqs_queue.demo: Creating...
aws_s3_bucket.demo: Creating...
aws_s3_bucket.demo: Creation complete after 2s [id=sukhmay-demo-bucket-2026-001]
aws_sqs_queue.demo: Still creating... [00m10s elapsed]
aws_sqs_queue.demo: Still creating... [00m20s elapsed]
aws_sqs_queue.demo: Creation complete after 26s
[id=https://sqs.ap-south-2.amazonaws.com/183192605590/demo-queue]
```

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

```
[terraform-user@ip-172-31-5-142 terraform-project]$ vi iam.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ mkdir lambda
[terraform-user@ip-172-31-5-142 terraform-project]$ vi lambda/function.py
[terraform-user@ip-172-31-5-142 terraform-project]$ cd lambda
zip function.zip function.py
cd ..
  adding: function.py (deflated 32%)
[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ vi lambda.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ vi lambda.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ vi s3_sqs.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ vi s3.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ vi s3.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ vi s3.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform fmt
s3_sqs.tf
[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform validate
```

Error: Reference to undeclared resource

```
on s3_sqs.tf line 14, in resource "aws_sqs_queue_policy" "allow_s3":
14:         "aws:SourceArn" = aws_s3_bucket.demo.arn
```

A managed resource "aws_s3_bucket" "demo" has not been declared in the root module.

```
[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ cat s3.tf
resource "aws_s3_bucket_notification" "notify" {

  bucket = aws_s3_bucket.demo.id

  queue {
    queue_arn = aws_sqs_queue.demo.arn
    events    = ["s3:ObjectCreated:*"]
  }

  depends_on = [aws_sqs_queue_policy.allow_s3]
}
[terraform-user@ip-172-31-5-142 terraform-project]$ Read from remote host
ec2-16-112-129-181.ap-south-2.compute.amazonaws.com: Connection reset by peer
Connection to ec2-16-112-129-181.ap-south-2.compute.amazonaws.com closed.
client_loop: send disconnect: Connection reset by peer
```

X

```
2026-06-20 18:25.35 /drives/c/Users/USER/Downloads ssh -i "Aws-project.pem"
ec2-user@ec2-16-112-129-181.ap-south-2.compute.amazonaws.com
X11 forwarding request failed on channel 0
```

```

,      #_
~\    #####_      Amazon Linux 2023
~\    #####\
~\    \###|
~\    \#/  _____
~      V~'  '->
~~~~
~~~~
~.._._/
~\_/
~/m/'

```

```
Last login: Sat Jun 20 12:34:13 2026 from 27.61.127.142
[ec2-user@ip-172-31-5-142 ~]$ sudo su
[root@ip-172-31-5-142 ec2-user]# cd
[root@ip-172-31-5-142 ~]# su - terraform-user
Last login: Sat Jun 20 12:34:35 UTC 2026 on pts/7
[terraform-user@ip-172-31-5-142 ~]$ ls
terraform-project
[terraform-user@ip-172-31-5-142 ~]$ cd terraformproject
-bash: cd: terraformproject: No such file or directory
[terraform-user@ip-172-31-5-142 ~]$ cd terraform-project
[terraform-user@ip-172-31-5-142 terraform-project]$ ls
ec2.tf iam.tf lambda lambda.tf provider.tf s3.tf s3_sqs.tf security.tf sqs.tf
terraform.tfstate terraform.tfstate.backup vpc.tf
[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ vi s3_bucket.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ ls
ec2.tf iam.tf lambda lambda.tf provider.tf s3.tf s3_bucket.tf s3_sqs.tf
security.tf sqs.tf terraform.tfstate terraform.tfstate.backup vpc.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ vi s3_sqs.tf
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform validate
```

Error: Unclosed configuration block

```
on s3_sqs.tf line 1, in resource "aws_sqs_queue_policy" "allow_s3":
  1: resource "aws_sqs_queue_policy" "allow_s3" {
```

There is no closing brace for this block before the end of the file. This may be caused by incorrect brace nesting elsewhere in this file.

```
[terraform-user@ip-172-31-5-142 terraform-project]$  
[terraform-user@ip-172-31-5-142 terraform-project]$ vi s3_sqs.tf  
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform fmt  
terraform validate  
s3_sqs.tf  
Success! The configuration is valid.
```

```
[terraform-user@ip-172-31-5-142 terraform-project]$  
[terraform-user@ip-172-31-5-142 terraform-project]$ Read from remote host  
ec2-16-112-129-181.ap-south-2.compute.amazonaws.com: Connection reset by peer  
Connection to ec2-16-112-129-181.ap-south-2.compute.amazonaws.com closed.  
client loop: send disconnect: Connection reset by peer
```



```

        {
            + Statement = [
                + {
                    + Action      = "sts:AssumeRole"
                    + Effect      = "Allow"
                    + Principal = {
                        + Service = "lambda.amazonaws.com"
                    }
                },
            ]
            + Version = "2012-10-17"
        }
    )
    + create_date      = (known after apply)
    + force_detach_policies = false
    + id               = (known after apply)
    + managed_policy_arns = (known after apply)
    + max_session_duration = 3600
    + name              = "lambda-s3-sqs-role"
    + name_prefix       = (known after apply)
    + path               = "/"
    + tags_all           = (known after apply)
    + unique_id          = (known after apply)

    + inline_policy (known after apply)
}

# aws_iam_role_policy_attachment.basic_logs will be created
+ resource "aws_iam_role_policy_attachment" "basic_logs" {
    + id          = (known after apply)
    + policy_arn =
"arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole"
    + role        = "lambda-s3-sqs-role"
}

# aws_iam_role_policy_attachment.sqs_access will be created
+ resource "aws_iam_role_policy_attachment" "sqs_access" {
    + id          = (known after apply)
    + policy_arn = "arn:aws:iam::aws:policy/AmazonSQSFullAccess"
    + role        = "lambda-s3-sqs-role"
}

# aws_lambda_event_source_mapping.sqs_trigger will be created
+ resource "aws_lambda_event_source_mapping" "sqs_trigger" {
    + arn              = (known after apply)
    + batch_size       = 1
    + enabled          = true
    + event_source_arn =
"arn:aws:sqs:ap-south-2:183192605590:demo-queue"
    + function_arn     = (known after apply)
    + function_name     = (known after apply)
    + id               = (known after apply)
    + last_modified     = (known after apply)
    + last_processing_result = (known after apply)
    + maximum_record_age_in_seconds = (known after apply)
}

```

```

+ maximum_retry_attempts      = (known after apply)
+ parallelization_factor      = (known after apply)
+ region                      = "ap-south-2"
+ state                      = (known after apply)
+ state_transition_reason     = (known after apply)
+ tags_all                   = (known after apply)
+ uuid                      = (known after apply)

+ amazon_managed_kafka_event_source_config (known after apply)

+ self_managed_kafka_event_source_config (known after apply)
}

# aws_lambda_function.demo will be created
+ resource "aws_lambda_function" "demo" {
  + architectures      = (known after apply)
  + arn                = (known after apply)
  + code_sha256        = (known after apply)
  + filename           = "lambda/function.zip"
  + function_name      = "s3-sqs-lambda"
  + handler            = "function.lambda_handler"
  + id                = (known after apply)
  + invoke_arn         = (known after apply)
  + last_modified      = (known after apply)
  + memory_size        = 128
  + package_type       = "Zip"
  + publish            = false
  + qualified_arn      = (known after apply)
  + qualified_invoke_arn = (known after apply)
  + region             = "ap-south-2"
  + reserved_concurrent_executions = -1
  + response_streaming_invoke_arn = (known after apply)
  + role               = (known after apply)
  + runtime            = "python3.9"
  + signing_job_arn    = (known after apply)
  + signing_profile_version_arn = (known after apply)
  + skip_destroy       = false
  + source_code_hash   =
    "InNF7rQjjjT4WQl2Tn2+r9iEDyCKD4gZDLQWlQl2PdA="
  + source_code_size   = (known after apply)
  + tags_all           = (known after apply)
  + timeout            = 10
  + version            = (known after apply)

  + ephemeral_storage (known after apply)

  + logging_config (known after apply)

  + tracing_config (known after apply)
}

# aws_s3_bucket_notification.notify will be created
+ resource "aws_s3_bucket_notification" "notify" {
  + bucket      = "sukhmay-demo-bucket-2026-001"
  + eventbridge = false

```

```

+ id          = (known after apply)
+ region      = "ap-south-2"

+ queue {
  + events = [
    + "s3:ObjectCreated:*",
  ]
  + id      = (known after apply)
  + queue_arn = "arn:aws:sqs:ap-south-2:183192605590:demo-queue"
}

# aws_sqs_queue_policy.allow_s3 will be created
+ resource "aws_sqs_queue_policy" "allow_s3" {
  + id          = (known after apply)
  + policy      = jsonencode(
    {
      + Statement = [
        + {
          + Action      = "SQS:SendMessage"
          + Condition = {
            + ArnEquals = {
              + "aws:SourceArn" =
"arn:aws:s3:::sukhmay-demo-bucket-2026-001"
            }
          }
          + Effect      = "Allow"
          + Principal    = "*"
          + Resource     = "arn:aws:sqs:ap-south-2:183192605590:demo-queue"
        },
      ]
      + Version      = "2012-10-17"
    }
  )
  + queue_url = "https://sqs.ap-south-2.amazonaws.com/183192605590/demo-queue"
  + region    = "ap-south-2"
}

```

Plan: 7 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```

[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform apply
aws_sqs_queue.demo: Refreshing state...
[id=https://sqs.ap-south-2.amazonaws.com/183192605590/demo-queue]
aws_s3_bucket.demo: Refreshing state... [id=sukhmay-demo-bucket-2026-001]
aws_vpc.main: Refreshing state... [id=vpc-0ed3dccc5d799fbb9]
aws_subnet.public: Refreshing state... [id=subnet-0caf4afde5ceb4dce]
aws_security_group.ec2_sg: Refreshing state... [id=sg-010fa4c1c81ca32a2]
aws_internet_gateway.igw: Refreshing state... [id=igw-0ce5164dc047e0a65]
aws_route_table.public: Refreshing state... [id=rtb-0e058852b8dad8c2d]

```

```
aws_instance.demo: Refreshing state... [id=i-0529f98074e14d755]
aws_route_table_association.public: Refreshing state...
[id=rtbassoc-04a813e5b7f3a67bf]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_iam_role.lambda_role will be created
+ resource "aws_iam_role" "lambda_role" {
  + arn                = (known after apply)
  + assume_role_policy = jsonencode(
    {
      + Statement = [
        + {
          + Action    = "sts:AssumeRole"
          + Effect    = "Allow"
          + Principal = {
            + Service = "lambda.amazonaws.com"
          }
        },
      ]
    }
  + Version           = "2012-10-17"
)
+ create_date        = (known after apply)
+ force_detach_policies = false
+ id                 = (known after apply)
+ managed_policy_arns = (known after apply)
+ max_session_duration = 3600
+ name                = "lambda-s3-sqs-role"
+ name_prefix        = (known after apply)
+ path                = "/"
+ tags_all            = (known after apply)
+ unique_id           = (known after apply)

+ inline_policy (known after apply)
}

# aws_iam_role_policy_attachment.basic_logs will be created
+ resource "aws_iam_role_policy_attachment" "basic_logs" {
  + id          = (known after apply)
  + policy_arn =
"arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole"
  + role        = "lambda-s3-sqs-role"
}

# aws_iam_role_policy_attachment.sqs_access will be created
+ resource "aws_iam_role_policy_attachment" "sqs_access" {
  + id          = (known after apply)
  + policy_arn = "arn:aws:iam::aws:policy/AmazonSQSFullAccess"
  + role        = "lambda-s3-sqs-role"
}
```

```

# aws_lambda_event_source_mapping.sqs_trigger will be created
+ resource "aws_lambda_event_source_mapping" "sqs_trigger" {
  + arn                = (known after apply)
  + batch_size         = 1
  + enabled            = true
  + event_source_arn   =
"arn:aws:sqs:ap-south-2:183192605590:demo-queue"
  + function_arn       = (known after apply)
  + function_name      = (known after apply)
  + id                 = (known after apply)
  + last_modified      = (known after apply)
  + last_processing_result = (known after apply)
  + maximum_record_age_in_seconds = (known after apply)
  + maximum_retry_attempts = (known after apply)
  + parallelization_factor = (known after apply)
  + region             = "ap-south-2"
  + state              = (known after apply)
  + state_transition_reason = (known after apply)
  + tags_all           = (known after apply)
  + uuid               = (known after apply)

  + amazon_managed_kafka_event_source_config (known after apply)

  + self_managed_kafka_event_source_config (known after apply)
}

# aws_lambda_function.demo will be created
+ resource "aws_lambda_function" "demo" {
  + architectures      = (known after apply)
  + arn                = (known after apply)
  + code_sha256        = (known after apply)
  + filename           = "lambda/function.zip"
  + function_name      = "s3-sqs-lambda"
  + handler            = "function.lambda_handler"
  + id                 = (known after apply)
  + invoke_arn         = (known after apply)
  + last_modified      = (known after apply)
  + memory_size        = 128
  + package_type       = "Zip"
  + publish            = false
  + qualified_arn       = (known after apply)
  + qualified_invoke_arn = (known after apply)
  + region             = "ap-south-2"
  + reserved_concurrent_executions = -1
  + response_streaming_invoke_arn = (known after apply)
  + role               = (known after apply)
  + runtime            = "python3.9"
  + signing_job_arn     = (known after apply)
  + signing_profile_version_arn = (known after apply)
  + skip_destroy       = false
  + source_code_hash    =
"InNF7rQjjjT4WQl2Tn2+r9iEDyCKD4gZDLQWIQl2PdA="
  + source_code_size    = (known after apply)
  + tags_all           = (known after apply)

```

```

+ timeout = 10
+ version = (known after apply)

+ ephemeral_storage (known after apply)

+ logging_config (known after apply)

+ tracing_config (known after apply)
}

# aws_s3_bucket_notification.notify will be created
+ resource "aws_s3_bucket_notification" "notify" {
  + bucket = "sukhmay-demo-bucket-2026-001"
  + eventbridge = false
  + id = (known after apply)
  + region = "ap-south-2"

  + queue {
    + events = [
      + "s3:ObjectCreated:*",
    ]
    + id = (known after apply)
    + queue_arn = "arn:aws:sqs:ap-south-2:183192605590:demo-queue"
  }
}

# aws_sqs_queue_policy.allow_s3 will be created
+ resource "aws_sqs_queue_policy" "allow_s3" {
  + id = (known after apply)
  + policy = jsonencode(
    {
      + Statement = [
        + {
          + Action = "SQS:SendMessage"
          + Condition = {
            + ArnEquals = {
              + "aws:SourceArn" =
"arn:aws:s3:::sukhmay-demo-bucket-2026-001"
            }
          }
          + Effect = "Allow"
          + Principal = "*"
          + Resource = "arn:aws:sqs:ap-south-2:183192605590:demo-queue"
        },
      ]
      + Version = "2012-10-17"
    }
  )
  + queue_url = "https://sqs.ap-south-2.amazonaws.com/183192605590/demo-queue"
  + region = "ap-south-2"
}

```

Plan: 7 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```
aws_sqs_queue_policy.allow_s3: Creating...
aws_iam_role.lambda_role: Creating...
aws_iam_role.lambda_role: Creation complete after 2s [id=lambda-s3-sqs-role]
aws_iam_role_policy_attachment.basic_logs: Creating...
aws_iam_role_policy_attachment.sqs_access: Creating...
aws_lambda_function.demo: Creating...
aws_iam_role_policy_attachment.sqs_access: Creation complete after 1s
[id=lambda-s3-sqs-role/arn:aws:iam::aws:policy/AmazonSQSFullAccess]
aws_iam_role_policy_attachment.basic_logs: Creation complete after 1s
[id=lambda-s3-sqs-role/arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole]
aws_sqs_queue_policy.allow_s3: Still creating... [00m10s elapsed]
aws_lambda_function.demo: Still creating... [00m10s elapsed]
aws_lambda_function.demo: Creation complete after 14s [id=s3-sqs-lambda]
aws_lambda_event_source_mapping.sqs_trigger: Creating...
aws_sqs_queue_policy.allow_s3: Still creating... [00m20s elapsed]
aws_sqs_queue_policy.allow_s3: Creation complete after 26s
[id=https://sqs.ap-south-2.amazonaws.com/183192605590/demo-queue]
aws_s3_bucket_notification.notify: Creating...
aws_lambda_event_source_mapping.sqs_trigger: Still creating... [00m10s elapsed]
aws_s3_bucket_notification.notify: Creation complete after 0s
[id=sukhmay-demo-bucket-2026-001]
aws_lambda_event_source_mapping.sqs_trigger: Still creating... [00m20s elapsed]
aws_lambda_event_source_mapping.sqs_trigger: Creation complete after 23s
[id=cff1bb94-fed9-4527-bc0b-b9577d07c234]
```

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

```
[terraform-user@ip-172-31-5-142 terraform-project]$ echo "hello this is the text file i
m uploading in the bucket " > test.txt
aws s3 cp test.txt s3://sukhmay-demo-bucket-2026-001
upload: ./test.txt to s3://sukhmay-demo-bucket-2026-001/test.txt
[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ Read from remote host
ec2-16-112-129-181.ap-south-2.compute.amazonaws.com: Connection reset by peer
Connection to ec2-16-112-129-181.ap-south-2.compute.amazonaws.com closed.
client_loop: send disconnect: Connection reset by peer
```

X

```
2026-06-20 18:59.17 /drives/c/Users/USER/Downloads ssh -i "Aws-project.pem"
ec2-user@ec2-16-112-129-181.ap-south-2.compute.amazonaws.com
X11 forwarding request failed on channel 0
```

```
,      #_
~\_    #####_      Amazon Linux 2023
~~ \_  #####\
~~   \###|
~~    \#/  _____ https://aws.amazon.com/linux/amazon-linux-2023
~~      V~'  '->
~~~
~~~. _  _/
```

```

    _/_/_/
   _/m/'_/_/
Last login: Sat Jun 20 13:14:17 2026 from 27.61.127.142
[ec2-user@ip-172-31-5-142 ~]$ sudo su
[root@ip-172-31-5-142 ec2-user]# cd
[root@ip-172-31-5-142 ~]# su - terraform-user
Last login: Sat Jun 20 13:15:16 UTC 2026 on pts/11
[terraform-user@ip-172-31-5-142 ~]$ cd terraform-project
[terraform-user@ip-172-31-5-142 terraform-project]$ aws s3 cp test.txt
s3://sukhmay-demo-bucket-2026-001
upload: ./test.txt to s3://sukhmay-demo-bucket-2026-001/test.txt
[terraform-user@ip-172-31-5-142 terraform-project]$ aws sqs receive-message --queue-url
https://sqs.ap-south-2.amazonaws.com/183192605590/demo-queue
[terraform-user@ip-172-31-5-142 terraform-project]$ aws logs describe-log-groups
{
  "logGroups": [
    {
      "logGroupName": "/aws/lambda/s3-sqs-lambda",
      "creationTime": 1781961541489,
      "metricFilterCount": 0,
      "arn":
"arn:aws:logs:ap-south-2:183192605590:log-group:/aws/lambda/s3-sqs-lambda:*",
      "storedBytes": 0,
      "logGroupClass": "STANDARD",
      "logGroupArn":
"arn:aws:logs:ap-south-2:183192605590:log-group:/aws/lambda/s3-sqs-lambda",
      "deletionProtectionEnabled": false
    },
    {
      "logGroupName": "/aws/lambda/s3-uploadnotifications",
      "creationTime": 1781194031870,
      "metricFilterCount": 0,
      "arn":
"arn:aws:logs:ap-south-2:183192605590:log-group:/aws/lambda/s3-uploadnotifications:*"
    },
    {
      "logGroupName": "/aws/lambda/s3-sqs-lambda",
      "creationTime": 1781961541489,
      "metricFilterCount": 0,
      "arn":
"arn:aws:logs:ap-south-2:183192605590:log-group:/aws/lambda/s3-sqs-lambda",
      "storedBytes": 460,
      "logGroupClass": "STANDARD",
      "logGroupArn":
"arn:aws:logs:ap-south-2:183192605590:log-group:/aws/lambda/s3-uploadnotifications",
      "deletionProtectionEnabled": false
    }
  ]
}
[terraform-user@ip-172-31-5-142 terraform-project]$ aws logs tail
/aws/lambda/s3-sqs-lambda --follow
2026-06-20T13:22:57.733000+00:00 2026/06/20/[$LATEST]45da5de3f4714c7e85238ffde6b2f71f
START RequestId: 8428545c-5093-5d54-bddd-36ecc6fc2240 Version: $LATEST
2026-06-20T13:22:57.735000+00:00 2026/06/20/[$LATEST]45da5de3f4714c7e85238ffde6b2f71f
Event received: {"Records": [{"messageId": "9a096127-3a3a-4fb5-abc8-785aa9cb91d7",
"receiptHandle":
"AQEBIwgKAXrXgSoiqspcRECNRnfKXqSfpUwmw61eWwQJIWUBabewFgGD3ja6zi3z0jiEAefEy9kC1BDqHxmM
ObbI4nNrLfPkD7fV0IHyl4wBopyRIpC81c75FUgPB4q0Isnj/+6z6J1oIzRgsu4VWyUBcjrKfG8GV+yORQFB1
iJi9Y/M4CHX0cTZ6UM0RAnUlNkJE+Z0dyKYTyY3D1n5oao/AlYe0oyqGaRwNYCL8540fLDL2W3ZN54TqJ1qim
LmKnDavSgYiVxczWblKJPGoqgeqvPJ+1d478upc5L1m0KjFBSH0aWGF/s00/jiE+KgEcrepE6+kv+6MFYApjU
lDFUt04nMgr1AntrHq6axCQTyVzMXP3hPqEKThvYrNU0rIPEfzj1Xi9IN0JV/3/E+3MM3BQ==", "body":

```

```
"{"Records":[{"eventVersion":"2.4","eventSource":"aws:s3","awsRegion":"ap-south-2","eventTime":"2026-06-20T13:22:56.628Z","eventName":"ObjectCreated:Put","userIdentity":{"principalId":"AWS:AIDASVJZBKOLH2JFVBSYY"},"requestParameters":{"sourceIPAddress":"16.112.129.181"},"responseElements":{"x-amz-request-id":"9SJS6TVTGARQ5CAS","x-amz-id-2":"1z/danZi5p8aDADocNsWkN5bynFnkVzzggKd0MaZdUIl3wvzTqudacLx4Hkcg8B70d3Mzkmh45LxnsiqdS+aBMdR3gFP8y0mlcnCKpozKnk="},"s3":{"s3SchemaVersion":"1.0","configurationId":"tf-s3-queue-20260620131850510700000001"},"bucket":{"name":"sukhmay-demo-bucket-2026-001","ownerIdentity":{"principalId":"A3EXRFJPAS8S4D"},"arn":"arn:aws:s3::sukhmay-demo-bucket-2026-001"},"object":{"key":"test.txt","size":58,"eTag":"c766d1b70b42437a7713262fbdcf0cb4"},"sequencer":"006A36943099573C0B"}]}","attributes":{"ApproximateReceiveCount":"1","SentTimestamp":"1781961777686","SenderId":"AROA2TFAJMARS6G6BYE6V:S3-PROD-END","ApproximateFirstReceiveTimestamp":"1781961777691"},"messageAttributes":{},"md5OfBody":"20b706dc5d78cb30ae01d05c3cd2e268","eventSource":"aws:sqs","eventSourceARN":"arn:aws:sqs:ap-south-2:183192605590:demo-queue","awsRegion":"ap-south-2"}]}
```

2026-06-20T13:22:57.735000+00:00 2026/06/20/[\$LATEST]45da5de3f4714c7e85238ffde6b2f71f
Message from SQS:

```
{"Records":[{"eventVersion":"2.4","eventSource":"aws:s3","awsRegion":"ap-south-2","eventTime":"2026-06-20T13:22:56.628Z","eventName":"ObjectCreated:Put","userIdentity":{"principalId":"AWS:AIDASVJZBKOLH2JFVBSYY"},"requestParameters":{"sourceIPAddress":"16.112.129.181"},"responseElements":{"x-amz-request-id":"9SJS6TVTGARQ5CAS","x-amz-id-2":"1z/danZi5p8aDADocNsWkN5bynFnkVzzggKd0MaZdUIl3wvzTqudacLx4Hkcg8B70d3Mzkmh45LxnsiqdS+aBMdR3gFP8y0mlcnCKpozKnk="},"s3":{"s3SchemaVersion":"1.0","configurationId":"tf-s3-queue-20260620131850510700000001"},"bucket":{"name":"sukhmay-demo-bucket-2026-001","ownerIdentity":{"principalId":"A3EXRFJPAS8S4D"},"arn":"arn:aws:s3::sukhmay-demo-bucket-2026-001"},"object":{"key":"test.txt","size":58,"eTag":"c766d1b70b42437a7713262fbdcf0cb4"},"sequencer":"006A36943099573C0B"}]}
```

2026-06-20T13:22:57.736000+00:00 2026/06/20/[\$LATEST]45da5de3f4714c7e85238ffde6b2f71f
END RequestId: 8428545c-5093-5d54-bddd-36ecc6fc2240

2026-06-20T13:22:57.737000+00:00 2026/06/20/[\$LATEST]45da5de3f4714c7e85238ffde6b2f71f
REPORT RequestId: 8428545c-5093-5d54-bddd-36ecc6fc2240 Duration: 2.74 ms Billed
Duration: 3 ms Memory Size: 128 MB Max Memory Used: 31 MB

2026-06-20T13:31:02.410000+00:00 2026/06/20/[\$LATEST]7e944dee5a4241ea9af25828b941dd66
INIT_START Runtime Version: python:3.9.v133 Runtime Version ARN:
arn:aws:lambda:ap-south-2::runtime:b46f7bc0f3da8071d1b824471f2c69c8766b756b827eb0455d2118c622ae7bcf

2026-06-20T13:31:02.482000+00:00 2026/06/20/[\$LATEST]7e944dee5a4241ea9af25828b941dd66
START RequestId: 654229e1-8622-5792-98b7-4b511e70790f Version: \$LATEST

2026-06-20T13:31:02.483000+00:00 2026/06/20/[\$LATEST]7e944dee5a4241ea9af25828b941dd66
Event received: {"Records": [{"messageId": "439bb27d-d5f0-4956-8eac-f55493e85dca",
"receiptHandle":

```
"AQEBj4KqXR5t6Fg3T5WjejHVxC+METvQH0fMt9HyqNzqi49Y+5RmEz1pvYskcQ0/5ksTVNYGF5yciPCfSa13  
QJTvmXRwX/BQ4xFwRn3e5qsbie3og/SSKGYAfegE65gFDaj5FTmzWe9kkj8LVilT4Nm/9zR34SejGs0ljue5p  
lkt9+spNeFpcBPCqPW3GKxGPEah4Qr4Ju0RTZoUUh/o/eJZyhnobh5hZDdA3to+OIfd4tPozOGVHCzEWgGDxP  
z4DHPQs1/bBM8mm9NlGd9yS6+87u07B0GXkypJ2spGLljtcQ10faQ9PT7c0apCLd0Yziw8FP+XiiV4LtZzYL  
k/RjYI3g5XqJyFGZBvZgP+M9qDfHo9hf/QcKkP2Emp7w4zKmcZvP6ljCu5JGxjlcY8yRqQA=", "body":  
{"Records":[{"eventVersion":"2.4","eventSource":"aws:s3","awsRegion":"ap-south-2","eventTime":"2026-06-20T13:31:01.506Z","eventName":"ObjectCreated:Put","userIdentity":{"principalId":"AWS:AIDASVJZBKOLH2JFVBSYY"},"requestParameters":{"sourceIPAddress":"16.112.129.181"},"responseElements":{"x-amz-request-id":"JEB7J6V96FXN4CNJ","x-amz-id-2":"QW9046AEN1gOWmMG41M4PK8u0cAOgnCQgNAPrTJu9vx9ocCx42dW2U6pNQt721I5xXTPzNvRpw2SswF9HNAHilEZlJLKhvps3QA+qhfy7Q="},"s3":{"s3SchemaVersion":"1.0","configurationId":"tf-s3-queue-20260620131850510700000001"},"bucket":{"name":"sukhmay-demo-bucket-2026-001","ownerIdentity":{"principal
```

```
Id\": \"A3EXRFJPAS8S4D\"}, \"arn\": \"arn:aws:s3::sukhmay-demo-bucket-2026-001\"}, \"object\": {\"key\": \"test.txt\", \"size\": 58, \"eTag\": \"c766d1b70b42437a7713262fbdcf0cb4\", \"sequencer\": \"006A3696157B059AE0\"}}}}\", \"attributes\": {\"ApproximateReceiveCount\": \"1\", \"SentTimestamp\": \"1781962262277\", \"SenderId\": \"AROA2TFAJMARS6G6BYE6V:S3-PROD-END\", \"ApproximateFirstReceiveTimestamp\": \"1781962262283\"}, \"messageAttributes\": {}, \"md5OfBody\": \"265add57c1bd70392b695f82e12840f8\", \"eventSource\": \"aws:sqs\", \"eventSourceARN\": \"arn:aws:sqs:ap-south-2:183192605590:demo-queue\", \"awsRegion\": \"ap-south-2\"}}}
```

2026-06-20T13:31:02.483000+00:00 2026/06/20/[\$LATEST]7e944dee5a4241ea9af25828b941dd66

Message from SQS:

```
{\"Records\": [{\"eventVersion\": \"2.4\", \"eventSource\": \"aws:s3\", \"awsRegion\": \"ap-south-2\", \"eventTime\": \"2026-06-20T13:31:01.506Z\", \"eventName\": \"ObjectCreated:Put\", \"userIdentity\": {\"principalId\": \"AWS:AIDASVJZBKOLH2JFVBSYY\"}, \"requestParameters\": {\"sourceIPAddress\": \"16.112.129.181\"}, \"responseElements\": {\"x-amz-request-id\": \"JEB7J6V96FXN4CNJ\", \"x-amz-id-2\": \"QW9046AEN1gOWmMG41M4PK8u0cAOgnCQgNAprTJu9vx9ocCx42dW2U6pNQ721I5xXTPzNvRpw2SswF9HNAHilEZlJLKhhvpS3QA+qhfY7Q=\"}, \"s3\": {\"s3SchemaVersion\": \"1.0\", \"configurationId\": \"tf-s3-queue-20260620131850510700000001\", \"bucket\": {\"name\": \"sukhmay-demo-bucket-2026-001\", \"ownerIdentity\": {\"principalId\": \"A3EXRFJPAS8S4D\"}, \"arn\": \"arn:aws:s3::sukhmay-demo-bucket-2026-001\"}, \"object\": {\"key\": \"test.txt\", \"size\": 58, \"eTag\": \"c766d1b70b42437a7713262fbdcf0cb4\", \"sequencer\": \"006A3696157B059AE0\"}}}}}
```

2026-06-20T13:31:02.484000+00:00 2026/06/20/[\$LATEST]7e944dee5a4241ea9af25828b941dd66
END RequestId: 654229e1-8622-5792-98b7-4b511e70790f

2026-06-20T13:31:02.484000+00:00 2026/06/20/[\$LATEST]7e944dee5a4241ea9af25828b941dd66
REPORT RequestId: 654229e1-8622-5792-98b7-4b511e70790f Duration: 1.33 ms Billed
Duration: 71 ms Memory Size: 128 MB Max Memory Used: 31 MB Init Duration: 69.31 ms

- MobaXterm Personal Edition v25.4 •
(X server, SSH client and network tools)

- Your computer drives are accessible through the `/drives` path
- Your DISPLAY is set to `10.38.111.199:0.0`
- When using `SSH`, your remote DISPLAY is automatically forwarded
- Each command status is specified by a special symbol (✓ or ✗)

- Important:

This is MobaXterm Personal Edition. The Professional edition allows you to customize MobaXterm for your company: you can add your own logo, your parameters, your welcome message and generate either an MSI installation package or a portable executable. We can also modify MobaXterm or develop the plugins you need.

For more information: <https://mobaxterm.mobatek.net/download.html>

```
2026-06-20 21:18:01 /drives/c/Users/USER/Downloads ssh -i "Aws-project.pem"
ec2-user@ec2-16-112-129-181.ap-south-2.compute.amazonaws.com
X11 forwarding request failed on channel 0
```

```

      , #_
    ~ \ ##### Amazon Linux 2023
    ~ \ #####\
    ~   \###|
    ~     \#/_ https://aws.amazon.com/linux/amazon-linux-2023
    ~       V~' '->
    ~~~
    ~ . _ . /
      -/_/_/_/
        /m/'

```

```
Last login: Sat Jun 20 13:29:45 2026 from 27.61.127.142
[ec2-user@ip-172-31-5-142 ~]$ sudo su
[root@ip-172-31-5-142 ec2-user]# cd
[root@ip-172-31-5-142 ~]# ld
ld: no input files
[root@ip-172-31-5-142 ~]# su - terraform-user
Last login: Sat Jun 20 13:30:33 UTC 2026 on pts/1
[terraform-user@ip-172-31-5-142 ~]$ cd terraform-project
[terraform-user@ip-172-31-5-142 terraform-project]$ lx
-bash: lx: command not found
[terraform-user@ip-172-31-5-142 terraform-project]$ ls
ec2.tf  lambda      provider.tf  s3_bucket.tf  security.tf  terraform.tfstate
test.txt
iam.tf  lambda.tf  s3.tf      s3_sqs.tf    sqs.tf      terraform.tfstate.backup
vpc.tf
[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ ls lambda
function.py  function.zip
[terraform-user@ip-172-31-5-142 terraform-project]$ cat lambda.tf
resource "aws_lambda_function" "demo" {

    function_name = "s3-sqs-lambda"
    role          = aws_iam_role.lambda_role.arn
```

```

    handler = "function.lambda_handler"
    runtime = "python3.9"

    filename          = "lambda/function.zip"
    source_code_hash = filebase64sha256("lambda/function.zip")

    timeout = 10
}

resource "aws_lambda_event_source_mapping" "sqs_trigger" {

    event_source_arn = aws_sqs_queue.demo.arn
    function_name    = aws_lambda_function.demo.arn

    batch_size = 1
}

[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ cat lambda/function.py
import json

def lambda_handler(event, context):
    print("Event received:", json.dumps(event))

    for record in event['Records']:
        body = record['body']
        print("Message from SQS:", body)

    return {
        'statusCode': 200
    }
[terraform-user@ip-172-31-5-142 terraform-project]$
[terraform-user@ip-172-31-5-142 terraform-project]$ vi lambda/function.py
[terraform-user@ip-172-31-5-142 terraform-project]$ cd lambda
rm function.zip
zip function.zip function.py
cd ..
  adding: function.py (deflated 46%)
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform plan
aws_iam_role.lambda_role: Refreshing state... [id=lambda-s3-sqs-role]
aws_sqs_queue.demo: Refreshing state...
[id=https://sqs.ap-south-2.amazonaws.com/183192605590/demo-queue]
aws_vpc.main: Refreshing state... [id=vpc-0ed3dccd5d799fbb9]
aws_s3_bucket.demo: Refreshing state... [id=sukhmay-demo-bucket-2026-001]
aws_sqs_queue_policy.allow_s3: Refreshing state...
[id=https://sqs.ap-south-2.amazonaws.com/183192605590/demo-queue]
aws_s3_bucket_notification.notify: Refreshing state...
[id=sukhmay-demo-bucket-2026-001]
aws_security_group.ec2_sg: Refreshing state... [id=sg-010fa4c1c81ca32a2]
aws_internet_gateway.igw: Refreshing state... [id=igw-0ce5164dc047e0a65]
aws_subnet.public: Refreshing state... [id=subnet-0caf4afde5ceb4dce]
aws_route_table.public: Refreshing state... [id=rtb-0e058852b8dad8c2d]
aws_route_table_association.public: Refreshing state...

```

```
[id=rtbassoc-04a813e5b7f3a67bf]
aws_instance.demo: Refreshing state... [id=i-0529f98074e14d755]
aws_iam_role_policy_attachment.basic_logs: Refreshing state...
[id=lambda-s3-sqs-role/arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole]
aws_iam_role_policy_attachment.sqs_access: Refreshing state...
[id=lambda-s3-sqs-role/arn:aws:iam::aws:policy/AmazonSQSFullAccess]
aws_lambda_function.demo: Refreshing state... [id=s3-sqs-lambda]
aws_lambda_event_source_mapping.sqs_trigger: Refreshing state...
[id=cff1bb94-fed9-4527-bc0b-b9577d07c234]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

~ update in-place

Terraform will perform the following actions:

```
# aws_lambda_function.demo will be updated in-place
~ resource "aws_lambda_function" "demo" {
  id                        = "s3-sqs-lambda"
  ~ last_modified          = "2026-06-20T13:18:35.409+0000" -> (known after
apply)
  ~ source_code_hash       =
  "InNF7rQjjjT4WQl2Tn2+r9iEDyCKD4gZDLQWIQl2PdA=" ->
  "UimG/wNoIBHIp4Ww6QMG0l9/aa8ma+BAvjNjyjAUvZY="
  tags                     = {}
  # (29 unchanged attributes hidden)

  # (3 unchanged blocks hidden)
}
```

Plan: 0 to add, 1 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```
[terraform-user@ip-172-31-5-142 terraform-project]$ terraform apply
aws_sqs_queue.demo: Refreshing state...
[id=https://sqs.ap-south-2.amazonaws.com/183192605590/demo-queue]
aws_vpc.main: Refreshing state... [id=vpc-0ed3dccd5d799fbb9]
aws_s3_bucket.demo: Refreshing state... [id=sukhmay-demo-bucket-2026-001]
aws_iam_role.lambda_role: Refreshing state... [id=lambda-s3-sqs-role]
aws_sqs_queue_policy.allow_s3: Refreshing state...
[id=https://sqs.ap-south-2.amazonaws.com/183192605590/demo-queue]
aws_s3_bucket_notification.notify: Refreshing state...
[id=sukhmay-demo-bucket-2026-001]
aws_security_group.ec2_sg: Refreshing state... [id=sg-010fa4c1c81ca32a2]
aws_subnet.public: Refreshing state... [id=subnet-0caf4afde5ceb4dce]
aws_internet_gateway.igw: Refreshing state... [id=igw-0ce5164dc047e0a65]
aws_route_table.public: Refreshing state... [id=rtb-0e058852b8dad8c2d]
aws_instance.demo: Refreshing state... [id=i-0529f98074e14d755]
aws_route_table_association.public: Refreshing state...
[id=rtbassoc-04a813e5b7f3a67bf]
```

```
aws_lambda_function.demo: Refreshing state... [id=s3-sqs-lambda]
aws_iam_role_policy_attachment.basic_logs: Refreshing state...
[id=lambda-s3-sqs-role/arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole]
aws_iam_role_policy_attachment.sqs_access: Refreshing state...
[id=lambda-s3-sqs-role/arn:aws:iam::aws:policy/AmazonSQSFullAccess]
aws_lambda_event_source_mapping.sqs_trigger: Refreshing state...
[id=cff1bb94-fed9-4527-bc0b-b9577d07c234]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

~ update in-place

Terraform will perform the following actions:

```
# aws_lambda_function.demo will be updated in-place
~ resource "aws_lambda_function" "demo" {
  id                        = "s3-sqs-lambda"
  ~ last_modified          = "2026-06-20T13:18:35.409+0000" -> (known after
apply)
  ~ source_code_hash       =
  "InNF7rQjjjT4WQl2Tn2+r9iEDyCKD4gZDLQWlQl2PdA=" ->
  "UimG/wNoIBHIp4Ww6QMG0l9/aa8ma+BAvjNjyJAUVZY="
  tags                    = {}
  # (29 unchanged attributes hidden)

  # (3 unchanged blocks hidden)
}
```

Plan: 0 to add, 1 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

```
aws_lambda_function.demo: Modifying... [id=s3-sqs-lambda]
aws_lambda_function.demo: Modifications complete after 5s [id=s3-sqs-lambda]
```

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.

```
[terraform-user@ip-172-31-5-142 terraform-project]$ echo "Hello this is my test3file
uploading in the s3 bucket" > test3.txt
aws s3 cp test3.txt s3://sukhmay-demo-bucket-2026-001
upload: ./test3.txt to s3://sukhmay-demo-bucket-2026-001/test3.txt
[terraform-user@ip-172-31-5-142 terraform-project]$ aws logs tail
/aws/lambda/s3-sqs-lambda --follow
2026-06-20T15:54:45.981000+00:00 2026/06/20/[$LATEST]069051c0e39a4b9d9767fe9942797fb7
INIT_START Runtime Version: python:3.9.v133 Runtime Version ARN:
arn:aws:lambda:ap-south-2::runtime:b46f7bc0f3da8071d1b824471f2c69c8766b756b827eb0455d
2118c622ae7bcf
2026-06-20T15:54:46.058000+00:00 2026/06/20/[$LATEST]069051c0e39a4b9d9767fe9942797fb7
START RequestId: 66d734eb-09d4-5903-b246-565bb0285766 Version: $LATEST
2026-06-20T15:54:46.059000+00:00 2026/06/20/[$LATEST]069051c0e39a4b9d9767fe9942797fb7
Event received: {"Records": [{"messageId": "1e734dd1-5749-4372-97c6-c386e5d33dae",
```

"receiptHandle":

```
"AQEBBoqN+aomtsb6FkVbPmX2QtWfPfHMyGLojslpdozpDdSr/ATSz3pLEsRQOK1I/rRjmW42qyuJYj2qpiRW
DKRLz88FcXxF2aiP/XGeLYC9C0CSS53kdYzF3vDjecU607vaWXg3J+0Mdb8ZL2X0fMilpGZI9Eii9Fy+QbIeR
aIGLYiAukQxovvND6jTqq9oAL2skphL0RyvR48VcWm3/MYgpF86FisEreqWNHbNTVr3aeJDPGNISJL4g9S90t
3buE5pw1Z7P0x4jono2cE8DNN6PK0QCVXy20SvFKqqb3TsbWouKtCcoRWLuhuyswB3w+hXi3ZJyaj9E+5QHUX
ImOGrrgDNM5EbUh83059+2RFLEmtqLhNmj5lsU1qd0+Y0vHd2WUzXGoZ+NAUHUYMN4tCeZw==", "body":
{"Records":[{"eventVersion":"2.4","eventSource":"aws:s3","awsRegion":"ap-south-2","eventTime":"2026-06-20T15:54:45.163Z","eventName":"ObjectCreated:
Put","userIdentity":{"principalId":"AWS:AIDASVJZBK0LH2JFVBSYY"},"requestParameters":{"sourceIPAddress":"16.112.129.181"},"responseElements":{"x-amz-reques
t-id":"6WX324S2BDQGRE3B","x-amz-id-2":"MjC60r+adVLUYg/eR9YzHU+BlTgimGyknUluRVws
sBPsgBm4gwkFaM8Q1AU7JyA0XdEq/nd9arPV8WUMDZ6eoKihfR1l6GH82E1HL+sY1JU="},"s3":{"s3S
chemaVersion":"1.0","configurationId":"tf-s3-queue-20260620131850510700000001"
,"bucket":{"name":"sukhmay-demo-bucket-2026-001","ownerIdentity":{"principal
Id":"A3EXRFJPAS8S4D"},"arn":"arn:aws:s3::sukhmay-demo-bucket-2026-001"},"obj
ect":{"key":"test3.txt","size":54,"eTag":"d4a38d34f31f061a37f5776b1f5546d1"
},"sequencer":"006A36B7C5238C23DC"}}}], "attributes":
{"ApproximateReceiveCount": "1", "SentTimestamp": "1781970885832", "SenderId":
"AROAZTFAJMARS6G6BYE6V:S3-PROD-END", "ApproximateFirstReceiveTimestamp":
"1781970885838"}, "messageAttributes": {}, "md5OfBody":
"ee0ca7cdb958d369de5b4d4a7f0bfb57", "eventSource": "aws:sqs", "eventSourceARN":
"arn:aws:sqs:ap-south-2:183192605590:demo-queue", "awsRegion": "ap-south-2"]}]
2026-06-20T15:54:46.059000+00:00 2026/06/20/[$LATEST]069051c0e39a4b9d9767fe9942797fb7
Object uploaded: test3.txt (bucket: sukhmay-demo-bucket-2026-001)
2026-06-20T15:54:46.060000+00:00 2026/06/20/[$LATEST]069051c0e39a4b9d9767fe9942797fb7
END RequestId: 66d734eb-09d4-5903-b246-565bb0285766
2026-06-20T15:54:46.060000+00:00 2026/06/20/[$LATEST]069051c0e39a4b9d9767fe9942797fb7
REPORT RequestId: 66d734eb-09d4-5903-b246-565bb0285766 Duration: 1.42 ms Billed
Duration: 75 ms Memory Size: 128 MB Max Memory Used: 31 MB Init Duration: 73.17 ms
[terraform-user@ip-172-31-5-142 terraform-project]$
```